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CERTIFICATE

This is to certify that the following students of M.Sc. Part-II, have successfully completed the project entitled

**“STATISTICAL MEASUREMENT OF SERVICE QUALITY
OF INDIAN AIRLINES”**

During the Academic year of 2023-2024 Students involved in this research group are

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This research project is to the best of our knowledge and belief is original.

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ABSTRACT

In the dynamic landscape of Indian domestic airlines, understanding and enhancing service quality is paramount for ensuring customer satisfaction and loyalty.

This study explores the service quality landscape of Indian domestic airlines through the lens of the SERVQUAL instrument—a widely utilized framework for assessing service quality perceptions across diverse industries. Leveraging a dataset comprising responses from 440 passengers, collected via a meticulously designed Google Form survey, our research delves into the intricacies of service quality across five dimensions: reliability, assurance, tangible, empathy, and responsiveness.

Employing advanced statistical techniques including log-linear analysis, decision tree analysis, cluster analysis, discriminant analysis, structural equation modeling, and multinomial logistic regression, we aim to uncover hidden patterns and discerning passenger preferences. The analysis encompasses 56 variables encapsulating various facets of the airline passenger experience, ranging from demographic attributes and booking behaviors to onboard amenities, staff interactions, cleanliness perceptions, and overall satisfaction metrics.

Key findings shed light on the critical determinants of service quality within the Indian domestic aviation sector, offering actionable insights for industry stakeholders. By aligning service offerings with customer expectations, addressing gaps in reliability, assurance, tangibles, empathy, and responsiveness, airlines can enhance passenger experiences, foster brand loyalty, and thrive in an increasingly competitive marketplace.

This research contributes to the ongoing discourse on airline service quality by providing empirical evidence and actionable insights for industry stakeholders. By addressing the multifaceted needs of passengers and refining service offerings accordingly, airlines can enhance customer experiences, foster brand loyalty, and maintain a competitive edge in the market.

Keywords: Service quality, Customer satisfaction, Loyalty, Passenger preferences, Competitive marketplace, Multifaceted needs

CHAPTER I: INTRODUCTION TO THE INDIAN DOMESTIC AVIATION SECTOR

1.1 INTRODUCTION

The role of services is well recognized in the developed economies of the world. Service sector gives significant contribution in the GDP of the country. It not only provides the large number of employments but also trade in services improves economic performance significantly in the country.

The services sector accounts for 54 percent of India's Gross Value Added (GVA). The services sector made a remarkable recovery in the fiscal year 2022-23, posting a Year-on-Year (YoY) growth rate of 8.4%. This is a significant improvement compared to the 7.8% contraction witnessed in the preceding financial year, as emphasized in the Economic Survey 2022-23 presented by the Union Minister for Finance & Corporate Affairs, Smt Nirmala Sitharaman in Parliament today.

The importance of quality to the effectiveness and competitiveness of these services is also majorly accepted. Service organizations need to stay ahead in today's world of cut-throat competition by offering quality services and the same implies to the airline industry.

The aviation sector is a consistently growing and the most vital sector in the economic progress of a nation. It serves a key role in transporting people or products from one location to another, especially during short and distant transportation. The airline industry is one of the major services industries that considerably add to the wealth of developed and developing nations. Generally, airline sector is classified into domestic (64% of air traffic) and international (36%) and in specific, airline services are categorised into intercontinental, continental, regional or domestic (Ashgate, 2011).

1.1.1 Airline Industry in India

Domestic airlines sector in India has gained the utmost importance in the recent decade because a comprehensive passenger-focused study on the Indian domestic airline industry could not be found in the literature. Further, the complexity of measuring survival strategies has gained a lot of attention and researchers have found it imperative to identify the different service and marketing mix dimensions that are needed to satisfy customers' expectations.

The Indian aviation sector is an important contributor to the country's economy, providing employment opportunities and generating revenue. The airline service industry in India has experienced significant growth over the years, with the aviation sector in India expected to become the third largest globally (AirAsia: The Indian Challenge, 2021). This growth can be attributed to various factors, including an increase in the number of travellers and a favourable investment climate.

The aviation market in India has witnessed a surge in the number of travellers, with 150 million passengers passing through its airports. By the year 2020, this number is expected to reach 450 million, highlighting the immense potential of the industry. This growth has attracted the attention of foreign airline operators, who see India as a promising opportunity for investment. The Indian government has also taken steps to address challenges in the aviation sector. They have revised the foreign direct investment policies, allowing foreign companies to own up to 49% stake in Indian carriers.

This has led to an influx of foreign investments and joint ventures in the Indian market. The Indian aviation industry has undergone a rigorous and tremendous transformation in a short term, inching it closer to becoming the largest in the world in near future.

The quality of service in the Indian aviation sector is a topic of great importance and concern (Bellizzi et al., 2020). Given the challenges faced by airlines and airports in China, service quality becomes crucial for achieving a competitive advantage. Furthermore, the competition in the airline industry is fierce and airlines must strive to understand and meet the needs of passengers in order to provide a great experience and high-quality services.

Understanding the factors that influence customer satisfaction in the Indian aviation sector is essential for airlines to develop effective strategies to retain customers and improve their overall performance. Overall, this study will contribute valuable insights and recommendations for the Indian aviation sector to improve service quality, exceed customer expectations, and gain a competitive edge in the industry.

1.2 NEED OF STUDY

The increasing presence of private participants in the airline industry has intensified competition, compelling airlines to prioritize customer satisfaction as a fundamental aspect of their business operations. Quality service provision is crucial for retaining customer loyalty, securing market share, and ultimately ensuring organizational profitability. While price remains a primary factor in

attracting customers, some airlines are leveraging service excellence to differentiate their offerings and gain a competitive advantage.

Customers' expectations evolve throughout their journey with an airline, making it challenging to measure service quality due to factors such as varying perceptions, service variability, and inseparability. In this context, customers' perceptions play a significant role in defining service quality within the airline industry. Despite its importance in customer retention and attraction, airlines often struggle to deliver the appropriate service attributes.

Organizations across industries, particularly in aviation, are increasingly recognizing that offering quality service is not optional but imperative for global relevance while maintaining local significance. Despite a vast body of literature on the airline industry, there's a notable absence of comprehensive research focusing specifically on service quality within the aviation sector, particularly in the context of Indian airlines. Comparative analyses between Indian airlines are also lacking.

Given these circumstances, the present study aims to fill this research gap by undertaking an in-depth exploration of service quality within the Indian airline industry especially passengers travelling from Mumbai. By conducting a comparative analysis, the study seeks to shed light on the nuances of service provision among different airlines operating in India. Through this research endeavor, the researcher aims to contribute valuable insights that can inform strategies for enhancing service quality and competitiveness within the Indian aviation market.

1.3 OBJECTIVES

The study has following objectives:

1. To study the association between four key demographic variables: Purchase ticket from (Tktbuy), Airlines used most frequently (Airline), likelihood to recommend the airline to others (Recommend), and likelihood to travel again (Purchase Again).
2. To study the impact of various service quality dimensions of Airline passengers on his/her future recommendation about Airlines.
3. To study the impact of various service quality dimensions of Airline passengers on his/her future likeliness to travel again about Airlines.
4. To study the psychometric analysis of air passengers.
5. To check the effect of socio-economic and demographic variables on passenger's choice to travel again.
6. To understand the emotions of passengers.

CHAPTER II: LITERATURE REVIEW: UNDERSTANDING PASSENGER EXPERIENCE

Mark Wever (2016) have discussed that there is no boundary to the airline industry, there are flights to all routes worldwide. The airlines often face a cost reduction in order to be highly competitive but on the other hand they need to retain customers, they need to attract customers, for such reason they created loyalty programs for customers where they ensure that they can win over the customers.

Wang and Park (2016) have empirically studied that assuming the Cabin service could be a top priority for the customers during air travel, when customers perceive the service quality given around the cabin service the evaluation that is made is important. That is because, the cabin service can be presented to the customer in a differentiated manner which could be a very important strategy of the airline. When a customer has a positive image about the airline it substantially influences customer loyalty.

Kos, Kukar, and Vegelj (2017) have researched upon the variables that positively affect satisfaction are the tangibles and personnel quality satisfaction, in a positive way influences the intention of the customers to purchase, repurchase and recommend it to others. The airlines staff of quality, images as a strong driver of the antecedents of customer satisfaction. It is said that the tourists who are much satisfied, who may be on a leisure trip definitely recommend others to visit or repeat their visits to the chosen destination.

Kubra and Orkun (2017) have studied that the quality of Living has changed and technology has given a makeover to the service industries and therefore the expectations that the customers have towards the airline industry has undergone change. The airline companies need to keep a tab on the expectations of customers about new markets, service quality and bring them to their advantage by meeting the customers' expectations. Customers are satisfied when the tangibles in an airline are in place, the customer satisfaction increases, it is about comfort and hygiene.

Empathy is one of the prime emotions that need to be embedded in the service industry especially in the airline companies which has a positive influence on the customer satisfaction.

Lin and Suk (2016) have emphasized on service quality, active assistance by the crew is expected trustworthiness is a very important factor brought on within the customers by the crew. It is very important to understand the customer's request in service quality, high quality of cabin services is expected by the customers along with comfort preferential services are given to frequent flyers.

The luggage fees that are obtained by the airline should be cheap, for overweight luggage and fast and accurate service in case of changing situation is to be prompt by the staff towards the customer, these are the various service quality measures that need to be taken as a part of service

quality to the passengers.

Kim and Cho (2016) give a new dimension by a link between job satisfaction of the crew in airlines and customer satisfaction of passengers, service quality depends on the crew who provide services and satisfied the customers can only branch out of the employees when they are satisfied with the job which is dependent on few factors such as the management, work life balance and the appreciation that is meted out to the crew which may be monetary or maybe verbal.

Khanh (2017) in his paper have proved empirically that for measuring service quality there are few determinants which have direct effect on customer satisfaction which is reservation, flight staff, the aircraft, baggage check, in inflight services there are ways to improve service quality and enhance customer satisfaction. it is learnt that between male and female passengers there is no difference of satisfaction but among monthly income and age groups and education groups there is difference in satisfaction. So, it's mainly depending on the purpose of travel by the passenger and with which the passenger has certain level of expectations and that is satisfied by the services there is high level of satisfaction among passengers who travel for business this indicates that on time performance is expected of the airlines.

Gambo (2016) study establishes that there is no relationship between customer satisfaction and the check in process though there is a significant relationship between customer satisfaction and reliability baggage handling responsiveness and inflight services. The staff need to be trained on keeping good relations with the customers they need to be proactive and ensure that the arrival and departure are done on time they need to be responsive enough and most importantly maintains safety and wellbeing of the passengers.

Rahim (2016) study that service quality is connected to survival and competition by airlines which helps in building a long-term relationship with customers. Perceived service quality has its influence on the satisfaction of the passenger. The airlines need to improve service quality to sustain an amount of differentiation in order to survive competitiveness in the airline industry. Market focused strategies need to be developed to meet the needs and expectations of the customers and to give a better service quality should be measured on a regular basis to keep tab of the interests' expectations and needs of the passengers to enhance customer satisfaction.

Zaid, Ansari and Jeddah (2015) discuss upon refund on online cancellation of ticket is one factor that customers are not satisfied with. When they face this particular issue, they are tempted to switch from the present airline to the other airlines. Therefore, online services should be taken seriously by the airlines. When there is entry of foreign airlines the monopoly status of a certain airline in their native is removed

Sendilvelan, Suresh and Balachandran (2017) have empirically proved that prioritizing the importance according to respondents' responsiveness is the most important after which is tangibility then empathy then assurance and then reliability. The customers are satisfied more with the tangibility followed by responsiveness, reliability, empathy and assurance. Male passengers have their priorities according to the importance of service quality where female passengers have given their top priority to tangibility then the rest of the four variables.

Sandada and Matibiri (2016) have empirically proved the service quality dimensions have a significant positive effect on satisfaction and customer loyalty indicating that the customers who are satisfied may continue to fly with the same airline in the next travel. They will also spread word of mouth in a positive way and they may not switch to another airline. Frequent flyer program has a positive effect on customer loyalty. Having only satisfied customers is not enough to have loyal customers company should go beyond customer satisfaction, excite the passengers to increase their patronage to the airline and stop the switching behaviour. The word of mouth is very powerful tool to spread a positive image about the airline and satisfied customers should be enrolled within frequent flyer programs such that they are not likely to switch to other Airlines. Wen (2012) have discussed about after picture of service failure should proceed with service recovery which winds and strengthens the satisfaction of the customers and their trust, which leads to repurchase intention and word of mouth. Timeline is very important in the process of service recovery where a certain time frame is needed to solve the problem and keeping the customers informed about the progress such that it has a positive impact on the judgment made by customers post purchase. When there is less interaction between the customers and employee during a service recovery the satisfaction of the customers is low and it would be the perception of the customer that the failure was not solved in an efficient manner but if the fare procedures were followed then the satisfaction levels are high with a great sense of trust and commitment among customers now these customers will rebuy in the future and share positive word of mouth with others.

CHAPTER III: METHODOLOGY: ANALYZING PASSENGER PREFERENCES

3.1 DATA SOURCE

3.1.1 Research Instrument

The instrument here used is a seven-point Likert scale, with 1 = highly dissatisfied, 2 = dissatisfied, 3 = slightly dissatisfied, 4 = neutral, 5 = slightly satisfied, 6 = satisfied & 7 = very satisfied.

The first section collected the demographic information of the respondent including age, gender, profession, Monthly Income, Residential Status, Highest Qualification, Purpose of Travel, Frequency of Travel, Class of Travel, Choice decided for travel by, Purchase Ticket From & Airlines Used most Frequently.

The second section collects information about the customer's perception on quality of facilities provided by airlines using the following variables: Alternatives Provided in Case of Flight Delay, Boarding Announcement and Procedure, Ease of Finding check-in Counter, Ease of Getting Ticket, Ease of Website Navigation at the time of Booking, Waiting times during Check-In, Cabin Crew Announcements, Cleanliness of Cabin, In-flight Entertainment, Leg room available on Seat, Food Serving Experience, Quantity of Food Given, Food and beverage Choices, Efficiency of Check in Staff, Friendliness of Staff, Grooming and Appearance of Staff, Network of destinations/Connectivity, Innovative marketing schemes like special pricing, holiday packages etc, Seating Comfort, Smooth Take-off/Landing, Special facilities of the Airlines for Children, the Elderly and the Handicapped, Staff Assistance at the time of Arrival, Staff Attitudes at the time of Check-In, Cleanliness of Check-in Counter, Cleanliness of Lounge, Cleanliness of Washrooms, Conditions of Aircraft Carrier, Internet / WIFI / Sky phone options during Flight, Lounge Comfort and Space, Luggage delivery times, Medical Facility available during Flight, Newspaper and Magazines service during Flight, Communication skills of Staff, Dissemination of Information in case of Delay of Flight, On-time Flights departure and arrivals, Personalized attention given to you, Post-service Relationship & Security of the Luggage

3.1.2 Sampling and Data collection

This study implemented convenience sampling, a methodology often chosen for its accessibility and practicality in selecting research participants. The primary methodological approach employed in this research was a questionnaire survey, strategically designed to fulfil the overarching objectives. This survey targeted individuals who had experienced air travel with any Indian airline at least once. This specific criterion ensured that respondents had relevant

experiences with airline services, enabling them to provide informed opinions on the quality of facilities offered.

The questionnaire comprised of two distinct sections to gather data. The initial segment focused on gathering demographic information from the respondents. This included details such as age, gender, income level, and other factors that could potentially influence perceptions of service quality. The second part of the questionnaire comprised of the participants' perceptions of various aspects of airline services, aiming to capture their subjective assessments of quality across different facets of the travel experience.

The study utilized multiple methods for administering the survey instrument. While face-to-face interactions were employed to gather data directly from respondents, an online administration option was also made available.

The questionnaire was sent through email and other social media platforms, along with the physical distribution of the questionnaire. In total 440 responses were collected for the study. The sample included individuals who live in Mumbai.

The profile of respondents on the basis of the demographic variables is illustrated in the tables below (Tables- 1 to 12)

Statistics	AGE			
	Below 25 years	25-40 years	Above 40 years	Total
Frequency	172	178	90	440
Percentage	39.1	40.5	20.5	100.00

Table 1: Age Groups of Respondents

Statistics	GENDER		
	Male	Female	Total
Frequency	230	210	440
Percentage	52.3	47.7	100.00

Table 2: Gender Profile of Respondents.

Statistics	PROFESSION					
	Public	Private	Government	Business	Others	Total
Frequency	52	122	73	57	136	440
Percentage	11.8	27.8	16.6	13.00	30.80	100.00

Table 3: Occupation Profile of Respondents.

Statistics	MONTHLY INCOME					
	Below 50,000	50,000-1,00,000	1,00,000-2,00,000	2,00,000-3,00,000	Above 3,00,000	Total
Frequency	275	59	47	25	34	440
Percentage	62.5	13.4	10.7	5.7	7.7	100.00

Table 4: Income Profile of Respondents.

Statistics	RESIDENTIAL STATUS			
	Own	Rent	Other	Total
Frequency	355	65	20	440
Percentage	80.6	14.8	4.6	100

Table 5: Residential Profile of Respondents

Statistics	HIGHEST QUALIFICATION							
	Secondary	Higher-Secondary	Bachelor's	Master's	Doctorate	Professional	Others	Total
Frequency	3	83	158	108	46	35	7	440
Percentage	0.7	18.9	36	24.4	10.5	8	1.6	100

Table 6: Qualification Profile of Respondents

Statistics	PURPOSE OF TRAVEL			
	Business	Personal	Leisure	Total
Frequency	45	82	313	440
Percentage	10.2	18.6	71.1	100

Table 7: Travel Purpose Profile of Respondents

Statistics	FREQUENCY OF TRAVEL				
	Rarely	Occasionally	Frequently	Very Frequently	Total
Frequency	292	98	30	20	440
Percentage	66.4	22.3	6.8	4.5	100

Table 8: Flying Frequency Profile of Respondents.

Statistics	CLASS OF TRAVEL				
	Economy	Premium Economy	Business	First Class	Total
Frequency	358	34	26	22	440
Percentage	81.4	7.7	5.9	5	100

Table 9: Travel Class Profile of Respondents

Statistics	TRAVEL CHOICE DECIDED BY			
	Self	Family-Friends	Employer	Total
Frequency	138	288	14	440
Percentage	31.4	65.5	3.1	100

Table 10: Travel Choice Profile of Respondents.

Statistics	TICKET PURCHASED FROM				
	Airline Website	Travel Website	Counter	Travel Agent	Total
Frequency	282	83	36	39	440
Percentage	64.1	18.9	8.2	8.8	100

Table 11: Ticket Purchase Profile of Respondents.

Statistics	AIRLINES USED FREQUENTLY							
	Air Asia India	Air India	Go First	IndiGo	SpiceJet	Vistara	Others	Total
Frequency	167	81	9	100	20	40	23	440
Percentage	38	18.4	2	22.7	4.5	9.1	5.2	100

Table 12: Frequently Used Airline Profile of Respondents.

3.2 TOOLS AND TECHNIQUES

3.2.1 Loglinear Analysis

3.2.1.1 Introduction

Loglinear analysis is a statistical technique used to examine the associations between categorical variables in contingency tables. It is particularly well-suited for exploring the relationships between multiple categorical variables and identifying significant patterns or interactions among them.

At its core, loglinear analysis relies on the principles of maximum likelihood estimation to fit a model to the observed frequencies in the contingency table. The analysis involves specifying a model that represents the expected frequencies of the observed data under the assumption of independence between variables. The model is then compared to the observed frequencies using likelihood ratio tests to assess its goodness of fit.

Loglinear analysis finds applications in various fields, including social sciences, marketing research, epidemiology, and more. It is commonly used to analyse survey data, market research data, and observational studies where categorical variables are of interest.

3.2.1.2 Assumptions of Loglinear analysis

- The observations are independent and random.
- Observed frequencies are normally distributed about expected frequencies over repeated samples. This is a good approximation if both (a) The expected frequencies are greater than or equal to 5 for 80 % or more of the categories and (b) All expected frequencies are greater than 1.
- The logarithm of the expected value of the response variable is a linear combination of the explanatory variables.

3.2.1.3 Key Concepts in Loglinear Analysis:

1. Contingency Tables: Loglinear analysis begins with the construction of contingency tables, which organize the frequencies of observations for multiple categorical variables. These tables serve as the basis for identifying patterns and associations between variables.
2. Main Effects and Interactions: Loglinear models allow for the estimation of main effects, representing the influence of individual variables on the outcome, as well as interactions, capturing the joint influence of multiple variables.
3. Model Fitting: The process of fitting a loglinear model involves specifying a set of parameters that describe the expected frequencies of observations under the model. These parameters are estimated using maximum likelihood estimation techniques.
4. Model Comparison: Loglinear analysis involves comparing different models to determine the best-fitting model for the data. This comparison is typically done using likelihood ratio tests, which assess the improvement in fit provided by each additional parameter.

3.2.2 Cluster Analysis

3.2.2.1 What is cluster analysis?

The process of grouping a set of physical or abstract objects into classes of similar objects is called clustering. A cluster is a collection of data objects that are similar to one another within the same cluster and are dissimilar to the objects in other clusters. A cluster of data objects can be treated collectively as one group and so may be considered as a form of data compression. Cluster analysis is an important human activity. Early in childhood, we learn how to distinguish between cats and dogs, or between animals and plants, by continuously improving subconscious clustering schemes. By automated clustering, we can identify dense and sparse regions in object space and, therefore, discover overall distribution patterns and interesting correlations among data attributes.

In cluster analysis, a-priori information is not available about the group or cluster membership for any of the objects; rather, it is suggested by the data itself (Malhotra, 2005). There are two major cluster analysis techniques- hierarchical (agglomeration clustering) and non-hierarchical (divisive clustering), the difference being that hierarchical methods develop a tree structure in determining approximate number of clusters and nonhierarchical do not. For agglomerative hierarchical clustering, SPSS groups cases into progressively larger clusters until all cases are in one large cluster (George and Mallery, 2006). SPSS computes the smallest average distance between all group pairs and combines the two groups that are closest. The procedure begins with as many clusters as there are cases. At first step, the two cases with the smallest distance between them are clustered. Then it computes distance once more and combines the two that are next closest. This process continues until all cases are grouped into one large cluster.

Cluster analysis has been widely used in numerous applications, including market research, pattern recognition, data analysis, and image processing. In business, clustering can help marketers discover distinct groups in their customer bases and characterize customer groups based on purchasing patterns. In biology, it can be used to derive plant and animal taxonomies, categorize genes with similar functionality, and gain insight into structures inherent in populations. Clustering may also help in the identification of areas of similar land use in an earth observation database and in the identification of groups of houses in a city according to house type, value, and geographic location, as well as the identification of groups of automobile. Insurance policy holders with a high average claim cost. It can also be used to help classify documents on the Web for information discovery. Clustering is also called data segmentation in some applications because clustering partitions large data sets into groups according to their similarity.

Clustering can also be used for outlier detection, where outliers (values that are “far away” from any cluster) may be more interesting than common cases. Applications of outlier detection include the detection of credit card fraud and the monitoring of criminal activities in electronic commerce. For example, exceptional cases in credit card transactions, such as very expensive and frequent purchases, may be of interest as possible fraudulent activity. As a data Mining function, cluster analysis can be used as a stand-alone tool to gain insight into the distribution of data, to observe the characteristics of each cluster, and to focus on a particular set of clusters for further analysis.

Alternatively, it may serve as a preprocessing step for other algorithms, such as characterization, attribute subset selection, and classification, which would then operate on the detected clusters and the selected attributes or features.

During clustering the distances between the clusters are calculated. Since each cluster can contain more than one object, there are few methods to calculate the distance between two clusters called linkage criteria. The common choices are:

- Complete linkage (also known as the furthest neighbor and as the maximum method). It uses the maximal distance between two elements (one in each cluster) that is the distance between farthest elements in two clusters.
- Single linkage (also known as the nearest neighbor and as the minimum method). This method uses the distance between the closest pair of elements from the clusters.
- Average linkage (also known as Unweighted or Weighted Pair Group Method with Arithmetic Mean, UPGMA or WPGMA, respectively). In this case, the distance is calculated as an average of the distances between the pairs of elements from two clusters.

K-Means clustering is a good general-purpose way to think about discovering groups in data, but there are several aspects of it that are unsatisfying. For one, it requires the user to specify the number of clusters in advance, or to perform some kind of post hoc selection. For another, the notion of what forms a group is very simple: a datum belongs to cluster k if it is closer to the k th center than it is to any other center. Third, K-Means is nondeterministic; the solution it finds will depend on the initialization and even good initialization algorithms such as K-Means++ have a randomized aspect. Finally, we might reasonably think that our data are more complicated than can be described by simple partitions. For example, we partition organisms into different species, but science has also developed a rich taxonomy of living things: kingdom, phylum, class, etc.

Hierarchical clustering is one framework for thinking about how to address these shortcomings. Hierarchical clustering constructs a (usually binary) tree over the data. The leaves are individual data items, while the root is a single cluster that contains all of the data. Between the root and the leaves are intermediate clusters that contain subsets of the data. The main idea of hierarchical clustering is to make “clusters of clusters” going upwards to construct a tree. There are two main conceptual approaches to forming such a tree. Hierarchical agglomerative clustering (HAC) starts at the bottom, with every datum in its own singleton cluster, and merges groups together. Divisive clustering starts with all of the data in one big group and then chops it up until every datum is in its own singleton group.

Assumption of Hierarchical cluster analysis: The distance or similarity measures used should be appropriate for the data analyzed. Also, you should include all relevant variables in your analysis. Omission of influential variables can result in a misleading solution. Because hierarchical cluster analysis is an exploratory method, results should be treated as tentative until they are confirmed with an independent sample.

Assumption of K-means cluster analysis: Distances are computed using simple Euclidean distance. If you want to use another distance or similarity measure, use the Hierarchical Cluster Analysis procedure. Scaling of variables is an important consideration. If your variables are measured on different scales (for example, one variable is expressed in dollars and another variable is expressed in years), your results may be misleading. In such cases, you should consider standardizing your variables before you perform the k -means cluster analysis (this task can be done in the Descriptives procedure). The procedure assumes that you have selected the appropriate number of clusters and that you have included all relevant variables. If you have chosen an inappropriate number of clusters or omitted important variables, your results may be misleading.

3.2.2.2 Advantages of Using Cluster Analysis:

- Ability to recognize groups of items that are similar.
- Increased accuracy in predicting future behavior based on previous behavior within groups.
- Accuracy in forecasting customer preferences and needs.
- Improved understanding of the dynamics within customer groups
- Improved marketing efforts.

3.2.2.3 Disdvantages of Using Cluster Analysis:

- Longer time-consuming for data collection.
- More computational power is needed, especially for collocated and frequent items.
- Cluster analysis may be less effective if you allow outliers (extreme observation) to influence the results of your sample collection.
- Cluster analysis can be used in many reports, including customer satisfaction, stock market analysis, and testing software.

3.2.3 Decision Tree

3.2.3.1 Introduction

Decision tree analysis is a powerful data mining technique used for both classification and regression tasks. It involves constructing a tree-like structure where each internal node represents a decision based on a specific feature, and each leaf node represents the outcome or prediction.

Decision trees are intuitive and easy to interpret, making them popular for both exploratory and predictive analysis tasks. Decision tree analysis operates based on the divide-and-conquer principle, recursively partitioning the data into subsets based on the values of different features. The process involves selecting the best feature to split the data at each node, typically based on criteria such as information gain, Gini impurity, or entropy. This recursive partitioning continues until a stopping criterion is met, such as reaching a maximum tree depth or minimum number of samples per node.

Decision tree analysis has diverse applications across various domains, including finance, healthcare, marketing, and manufacturing. It is commonly used for customer segmentation, risk assessment, fraud detection, medical diagnosis, and more.

3.2.3.2 Key Concepts in Decision Tree Analysis

1. **Splitting Criteria:** Decision trees use various criteria to determine the best feature and split point for partitioning the data at each node. Common splitting criteria include information gain (for classification tasks), mean squared error reduction (for regression tasks), Gini impurity, and entropy.
2. **Pruning:** Decision trees are prone to overfitting, especially when they grow too deep and capture noise in the data. Pruning techniques are used to simplify and regularize the tree by removing unnecessary nodes and branches, thereby improving its generalization performance on unseen data.
3. **Tree Interpretation:** One of the key advantages of decision trees is their interpretability. The resulting tree structure can be easily visualized and understood, allowing users to interpret the decision-making process and identify important features driving the predictions.
4. **Ensemble Methods:** Decision trees can be enhanced through ensemble methods such as random forests and gradient boosting, which combine multiple trees to improve prediction accuracy and robustness.

3.2.4 Logistic Regression

3.2.4.1 Introduction

Logistic regression is a statistical analysis technique used to model the relationship between a binary dependent variable and one or more independent variables. The purpose of logistic regression is to estimate the probability of the dependent variable given a set of independent variables. We used binary logistic regression to find out which variables have a significant effect on binary dependent variable and thereby identified the risk factors. We express the formula for regression model in terms of the logit. Logit is the natural logarithm of the odds of outcome.

Odds is a numerical expression used to represent the probability of an event occurring and is expressed as the number of favorable to the number of non-favorable outcomes.

$$\text{logit} = L_i = B_0 + B_1X_1 + \dots + B_KX_K$$

where each X_i is a predictor and each B_i 's are the regression coefficients.

The coefficients B in logistic regression represent the change in the log odds of the outcome for a one-unit change in the corresponding independent variable, holding all other variables constant. The B coefficients with 0 indicate that the given explanatory variable does not affect the logit (that is, makes no difference in the probability of the dependent value equalling the value of the event, usually 1); positive or negative B coefficients indicate the explanatory variable increases or decreases the logit of the dependent.

An Odds > 1 means the independent variable increases odds (event). An Odds < 1 , then the independent variable decreases odds (event).

Logistic regression is widely used in various fields such as finance (credit scoring), healthcare (disease diagnosis), marketing (customer churn prediction), and more, where binary classification tasks are common.

3.2.4.2 Assumptions:

- The outcome variable should be binary, meaning it has only two possible classes or categories.
- The observations in the dataset are assumed to be independent of each other. This means that the value of one observation does not influence the value of another observation.
- The relationship between the independent variables and the log odds of the outcome should be linear. However, the relationship between the independent variables and the probabilities does not need to be linear.
- There should be little or no multicollinearity among the independent variables, meaning that the independent variables should not be highly correlated with each other.
- Logistic regression typically requires a relatively large sample size to produce stable and reliable estimates.

3.2.4.3 Multinomial logistic regression

One common extension of logistic regression is multinomial logistic regression, which allows for modelling scenarios where the outcome variable has more than two categories. Here's an overview of multinomial logistic regression: Instead of modelling the probability of a binary outcome, multinomial logistic regression models the probabilities of each category of the outcome variable. Coefficients in multinomial logistic regression represent the effect of each

independent variable on the log odds of being in a specific category compared to a reference category (usually chosen arbitrarily).

Multinomial logistic regression is applied in various fields where classification with more than two categories is required, such as sentiment analysis (classifying text into positive, negative, or neutral), disease diagnosis with multiple disease types, or customer segmentation into multiple segments.

3.2.4.4 Ordinal logistic regression

Ordinal logistic regression, also known as ordered logistic regression, is a statistical method used for modelling ordinal outcome variables. An ordinal outcome variable is one that has a natural order but does not have evenly spaced intervals between categories. Examples include Likert scale responses (e.g., strongly disagree, disagree, neutral, agree, strongly agree) or socioeconomic status categories (e.g., low, medium, high). In ordinal logistic regression, the cumulative probabilities of an observation falling into each category up to and including a particular category are modelled. Ordinal logistic regression finds applications in various fields, including social sciences (e.g., survey data analysis), psychology (e.g., attitude measurement), and healthcare (e.g., patient outcome prediction).

Ordinal logistic regression estimates two sets of parameters:

- Threshold parameters (α): These represent the intercepts for each category except for the last category (which serves as the reference category).
- Coefficients (β): These represent the effects of predictor variables on the log odds of an observation being in a higher category compared to the reference category.

Proportional odds assumption: This assumption states that the relationship between each pair of outcome categories and the predictor variables is constant across all categories. In other words, the effect of predictor variables on the odds of being in a higher category versus a lower category is the same regardless of the category being considered.

3.2.5 Sentiment Analysis

Sentiment analysis is the process of classifying whether a block of text is positive, negative, or neutral. It refers to the methods and strategies that enable firms to examine data about how their customer base feels about a given service or product. The goal that Sentiment mining tries to gain is to be analysed people's opinions in a way that can help businesses expand. It focuses not only on polarity (positive, negative & neutral) but also on emotions (happy, sad, angry, etc.). It uses various Natural Language Processing algorithms such as Rule-based, Automatic, and Hybrid.

Sentiment analysis is incredibly significant since it helps businesses understand their consumers sentiment towards their brand. By automatically classifying the emotions behind social media interactions, reviews, and more, organizations can make informed decisions. According to the survey, 80% of the world's data is unstructured. The data needs to be analysed and be in a structured manner whether it is in the form of emails, texts, documents, articles, and many more.

Pre-processing: Starting with collecting the text data that needs to be analysed for sentiment like customer reviews, social media posts, news articles, or any other form of textual content.

The collected text is pre-processed to clean and standardize the data with various tasks:

- Removing irrelevant information (e.g., HTML tags, special characters).
- Tokenization: Breaking the text into individual words or tokens.
- Removing stop words (common words like “and,” “the,” etc. that don't contribute much to sentiment).
- Stemming or Lemmatization: Reducing words to their root form.

Analysis: Text is converted for analysis using techniques like bag-of-words or word embeddings (e.g., Word2Vec, GloVe). Models are then trained with labelled datasets, associating text with sentiments (positive, negative, or neutral). After training and validation, the model predicts sentiment on new data, assigning labels based on learned patterns.

3.4.5.1 Challenges:

There are major challenges in the sentiment analysis approach:

- If the data is in the form of a tone, then it becomes really difficult to detect whether the comment is pessimist or optimistic.
- If the data is in the form of emoji, then you need to detect whether it is good or bad.
- Even the ironic, sarcastic, comparing comments detection is really hard.
- Comparing a neutral statement is a big task.

CHAPTER IV: DEMOGRAPHIC ANALYSIS: UNDERSTANDING CUSTOMER LOYALTY IN THE INDIAN DOMESTIC AVIATION SECTOR

4.1 INTRODUCTION

In this chapter, we delve into the intricate interplay between demographic factors and customer loyalty within the Indian Domestic aviation sector. With a focus on Understanding the drivers of travel behaviour and Loyalty among customers, we employ loglinear analysis to explore the associations between key demographic variables and indicators of customer loyalty.

The Indian domestic aviation sector is characterized by a diverse customer base with varying preferences, behaviours, and loyalties. As airlines vie for market share and seek to enhance customer satisfaction and retention, it becomes imperative to unravel the underlying demographic influences shaping customer perceptions and loyalty dynamics.

4.2 OBJECTIVES OF THE STUDY

- Our primary objective is to investigate the demographic factors that influence the travel and loyalty of customers towards Indian domestic airlines.
- Specifically, we aim to study the association between four key demographic variables: Purchase ticket from (Tktbuy), Airlines used most frequently (Airline), likelihood to recommend the airline to others (Recommend), and likelihood to travel again (Purchase Again).

4.3 RATIONALE FOR LOGLINEAR ANALYSIS

- Loglinear analysis offers a powerful framework for examining the complex relationships between categorical variables, making it particularly well-suited for exploring associations among demographic factors and customer loyalty indicators.
- By analysing contingency tables and estimating loglinear models, we can uncover significant patterns and interactions among the demographic variables of interest, shedding light on the drivers of customer loyalty in the Indian domestic aviation context.

4.4 SIGNIFICANCE OF THE STUDY

- Understanding the demographic influences on customer loyalty is essential for airlines seeking to tailor their marketing strategies, service offerings, and customer engagement initiatives to different segments of the market.
- By identifying demographic segments with distinct travel preferences and loyalty behaviours, airlines can optimize their resources and efforts to enhance customer satisfaction, loyalty, and advocacy.

4.5 METHODOLOGY

4.5.1 Data Collection

We collected primary data through a structured survey administered to passengers traveling within the Indian domestic aviation sector as well as with the use of Google form. The survey captured a comprehensive set of service quality variables, including aspects related to booking, check-in, onboard experience, staff interactions, cleanliness perceptions, and overall satisfaction ratings.

A total of 440 respondents participated in the survey, providing valuable insights into their preferences, satisfaction levels, and loyalty behaviours.

4.5.2 Variable Selection

From the set of 56 variables representing various dimensions of service quality, passenger demographics, and travel behaviour, we identified a set of four demographic variables. These variables were selected based on their relevance to the objectives of the study and their potential to influence travel and loyalty of customers.

4.5.3 Software Utilization

In this section, we delineate the software utilized in our study, namely IBM SPSS, to conduct Loglinear analysis and subsequent interpretation of results.

IBM SPSS:

- IBM SPSS (Statistical Package for the Social Sciences) was employed for data management, descriptive statistics, and preliminary analysis of the dataset.
- We utilized SPSS to perform data cleaning, data transformation, and variable recoding to prepare the dataset for decision tree analysis.

- Descriptive statistics such as frequencies, means, and standard deviations were calculated using SPSS to gain insights into the distribution and characteristics of the variables under investigation.
- Furthermore, SPSS facilitated the visualization of data through charts, histograms, and frequency tables, aiding in the exploration of patterns and trends in the dataset.

Integration of IBM SPSS:

- While IBM SPSS was primarily used for data management and initial exploratory analysis, and statistical analysis, including decision tree modelling.
- The integration of IBM SPSS allowed us to leverage the strengths of this software package, with its user-friendly interface and data manipulation capabilities.
- Data is processed and cleaned in SPSS and also analysed using decision tree analysis with greater precision, accuracy, validation.
- The results generated by Both SPSS with a precise and systematic approach are used for further interpretation and reporting.

4.5.4 Loglinear Analysis

Software Utilization:

In this study SPSS (Statistical Package for the Social Sciences) was employed for the loglinear analysis in this study. In SPSS, loglinear analysis was performed using the "Crosstabs" procedure, which allows for the creation of contingency tables and the estimation of loglinear models. The "Loglinear" option within the Crosstabs procedure enables the fitting of loglinear models to the data, allowing for the testing of associations between variables while controlling for other variables in the model.

The variables of interest in this study include Purchase ticket from (Tktbuy), Airlines used most frequently (Airline), likelihood to recommend the airline to others (Recommend), and likelihood to travel again (Purchase Again). These variables were treated as categorical and were included in the loglinear models to assess their associations with each other.

Loglinear models were estimated to test for main effects and interaction effects among the demographic variables and indicators of customer loyalty. The significance of associations was assessed using likelihood ratio tests and chi-square statistics, with p-values indicating the strength of evidence against the null hypothesis of no association.

Comparison and Validation:

The results of the loglinear analysis were interpreted to identify significant patterns and interactions among the demographic variables and indicators of customer loyalty. Insights gained

from the analysis provide valuable information for understanding the drivers of customer behaviour and preferences in the Indian domestic aviation sector.

4.5.5 Interpretation of results

The results of the loglinear analysis provide valuable insights into the associations between demographic variables and indicators of customer loyalty within the Indian domestic aviation sector. Several key findings emerged from the analysis, shedding light on the factors influencing customer behaviour and preferences:

1. **Significant Associations:** The loglinear model revealed significant associations between demographic variables such as Purchase ticket from (Tktbuy) and Airlines used most frequently (Airline), and indicators of customer loyalty including likelihood to recommend the airline to others (Recommend) and likelihood to travel again (Purchase Again).
2. **Interaction Effects:** Interaction effects between demographic variables were observed, indicating that the influence of one demographic factor on customer loyalty may vary depending on other demographic characteristics. For example, the association between Purchase ticket from (Tktbuy) and likelihood to recommend the airline to others (Recommend) may differ based on the Airlines used most frequently (Airline) by the passengers.
3. **Segmentation Patterns:** The analysis identified distinct segments of customers with different travel behaviours and loyalty preferences. By examining the joint distributions of demographic variables and customer loyalty indicators, meaningful segmentation patterns were uncovered, providing insights into the diverse needs and preferences of passengers.
4. **Implications for Airlines:** The insights gained from the analysis have implications for airlines operating in the Indian domestic aviation sector. Understanding the demographic factors that influence customer loyalty can inform strategic decision-making processes related to marketing, customer service, and product offerings. Airlines can tailor their services and promotions to better meet the needs and preferences of different customer segments, ultimately enhancing customer satisfaction and loyalty.

4.5.6 Limitations

While the loglinear analysis provides valuable insights into the associations between demographic variables and customer loyalty indicators, several limitations should be considered:

1. **Sample Representativeness:** The findings of the study may be influenced by the characteristics of the sample population and may not be generalizable to all passengers in the Indian domestic aviation sector. Efforts were made to ensure a diverse and representative sample, but biases may still exist.
2. **Data Limitations:** The analysis relies on self-reported survey data, which may be subject to recall bias and social desirability bias. Additionally, the scope of the survey may not capture all relevant factors influencing customer loyalty, leading to potential measurement errors.
3. **Model Assumptions:** Loglinear models make certain assumptions about the independence of observations and the linearity of relationships between variables. Violations of these assumptions may affect the validity of the results and interpretations.
4. **Causal Inference:** While the analysis identifies associations between demographic variables and customer loyalty indicators, it does not establish causal relationships. Other unobserved factors may confound the associations observed in the analysis.

4.5.7 Conclusions

In conclusion, the loglinear analysis conducted in this study provides valuable insights into the demographic influences on customer loyalty in the Indian domestic aviation sector. By examining the associations between demographic variables and indicators of customer loyalty, we gain a deeper understanding of the factors driving passenger behaviour and preferences. Despite the limitations inherent in the analysis, the findings offer actionable insights for airlines seeking to enhance customer satisfaction and loyalty. By tailoring their services and marketing efforts to different demographic segments, airlines can better meet the needs and preferences of their customers, ultimately fostering long-term relationships and sustainable growth in the competitive aviation market.

4.6 ANALYSIS

Variable Name	Categories
Purchase Ticket From (Tktbuy)	Airline website, Travel website, Counter, Travel agent
Airlines used most Frequently (Airline)	Air Asia India, Air India, Go First, IndiGo, SpiceJet, Vistara, Others
Recommendation (Recommend)	Yes, No
Purchase Again (Again)	Yes, No

Table 13: Variables and Categories of Loglinear Analysis

Interaction of 4 terms	Interaction of 3 terms	Interaction of 2 terms	Main Effect
Tktbuy * Airline * Again * Recommend	Tktbuy * Airline * Again	Tktbuy * Airline	Tktbuy
	Tktbuy * Airline * Recommend	Again * Recommend	Airline
	Tktbuy * Again * Recommend	Tktbuy * Recommend	Again
	Airline * Again * Recommend	Airline * Recommend	Recommend
		Tktbuy * Again	
		Airline * Again	
Objective: To retain minimum number of significant terms			

Table 14: Generalised Saturated Model

Table 13 shows the four demographic variables used in the analysis as well as their categories in detail. Similarly, Table 14 shows a short preview of the results that will be obtained after the Loglinear analysis.

4.6.1 Loglinear Analysis

Decision Criteria:

- when $p < 0.05$; Retain this term in the model.
- when $p > 0.05$; Remove this term in the model.

K	Main effects and Interaction effects	df	Likelihood Ratio		Pearson	
			Chi-Square	Sig.	Chi-Square	Sig.
1	i) Tktbuy ii) Airline iii) Again iv) Recommend	111	1764.147	.000	7329.745	.000
2	i)Tktbuy * Airline ii) Again * Recommend iii) Tktbuy * Recommend	100	1127.047	.000	1699.319	.000

	iv) Airline * Recommend V) Tktbuy * Again vi) Airline * Again					
3	i) Tktbuy * Airline * Again ii) Tktbuy * Airline * Recommend iii) Tktbuy * Again * Recommend iv) Airline * Again * Recommend	63	85.501	.031	205.043	.000
4	Tktbuy * Airline * Again * Recommend	18	1.353	1.000	.742	1.000
df used for these tests have NOT been adjusted for structural or sampling zeros. Tests using these df may be conservative.						
a. Tests that k-way and higher order effects are zero.						

Table 15: Saturated Model for Loglinear Analysis

4.6.2 Results of Loglinear Analysis

The loglinear analysis conducted in SPSS shown in Table 15 revealed several significant main effects and 2k interaction effects between demographic variables and indicators of customer loyalty within the Indian domestic aviation sector. The following key findings emerged from the analysis:

1. Significant Main Effects:

- Main effects refer to the individual effects of each demographic variable on customer loyalty indicators. In our analysis, several main effects were found to be significant, indicating that certain demographic factors independently influence customer loyalty.
- For example, the variables Purchase ticket from (Tktbuy) and Airlines used most frequently (Airline) showed significant main effects on likelihood to recommend the airline to others (Recommend) and likelihood to travel again (Purchase Again).

- Passengers who purchased tickets from specific channels or frequently used particular airlines were more likely to express loyalty towards the airline, as evidenced by their higher likelihood to recommend and repurchase.

2. Significant 2k Interaction Effects:

- Interaction effects refer to the combined effects of two or more demographic variables on customer loyalty indicators. In our analysis, significant 2k interaction effects were observed, indicating that the influence of one demographic factor on customer loyalty may be moderated by another demographic characteristic.
- For example, the interaction effect between Purchase ticket from (Tktbuy) and Airlines used most frequently (Airline) may influence the likelihood to recommend the airline to others (Recommend). The association between purchase behaviour and loyalty may vary depending on the airlines preferred by passengers.

4.6.3 Interpretation of Results

The significant main effects and 2k interaction effects observed in the loglinear analysis provide valuable insights into the demographic influences on customer loyalty in the Indian domestic aviation sector. By identifying these effects, we gain a deeper understanding of the factors driving passenger behaviour and preferences.

These findings highlight the importance of considering demographic factors in airline marketing and service strategies. Understanding how factors such as ticket purchase behaviour and airline preferences influence customer loyalty can help airlines tailor their services and promotions to better meet the needs and preferences of different customer segments.

4.6.4 Conclusions

In conclusion, the significant main effects and 2k interaction effects identified in the loglinear analysis offer valuable insights for airlines seeking to enhance customer satisfaction and loyalty. By understanding the demographic influences on customer behaviour, airlines can develop targeted strategies to improve customer engagement and foster long-term relationships.

These insights provide airlines with actionable information to optimize their marketing efforts, personalize their services, and enhance the overall customer experience. By addressing the specific needs and preferences of different customer segments, airlines can position themselves for success in the competitive Indian domestic aviation market.

CHAPTER V: PASSENGER PREFERENCES: INSIGHTS FROM CLUSTER ANALYSIS

5.1 VARIABLES USED :

To study the travel preferences and purchasing patterns of passengers within domestic airlines, the study included 38 service quality variables. The variables were:

Alternatives Provided in Case of Flight Delay, Boarding Announcement and Procedure, Ease of Finding check-in Counter, Ease of Getting Ticket, Ease of Website Navigation at the time of Booking, Waiting times during Check-In, Cabin Crew Announcements, Cleanliness of Cabin, In-flight Entertainment, Leg room available on Seat, Food Serving Experience, Quantity of Food Given, Food and beverage Choices, Efficiency of Check in Staff, Friendliness of Staff, Grooming and Appearance of Staff, Network of destinations/Connectivity, Innovative marketing schemes like special pricing, holiday packages etc, Seating Comfort, Smooth Take-off/Landing, Special facilities of the Airlines for Children, the Elderly and the Handicapped, Staff Assistance at the time of Arrival, Staff Attitudes at the time of Check-In, Cleanliness of Check-in Counter, Cleanliness of Lounge, Cleanliness of Washrooms, Conditions of Aircraft Carrier, Internet / WIFI / Sky phone options during Flight, Lounge Comfort and Space, Luggage delivery times, Medical Facility available during Flight, Newspaper and Magazines service during Flight, Communication skills of Staff, Dissemination of Information in case of Delay of Flight, On-time Flights departure and arrivals, Personalized attention given to you, Post-service Relationship & Security of the Luggage

Hierarchical clustering was carried out on the data to find the approximate number of clusters by using SPSS 26.0, displaying statistics outlined as agglomeration schedule, distance (or similarity) matrix, and cluster membership for a range of solutions. A non-hierarchical clustering method with k-mean clustering was conducted for a pre-determined final cluster centre (Nargundkar, 2007).

5.2 RESULTS

A reliability test was conducted on the scale. The statistics of reliability tests shows Cronbach's Alpha value of 0.985 which indicates an exceptionally high level of internal consistency among the items in the scale, this high level of internal consistency suggests that the scale is likely to produce reliable and stable measurements of whatever construct it is intended to assess. Analysis of variance (ANOVA) shows that the variables are significant at 95% confidence level ($F= 8.208$, p

= 0.0001). Thus, the final cluster Centres which describe the mean value of each variable for each of the clusters obtained for three cluster solution and the results are given below

The agglomeration schedule indicated formation of three clusters. K-mean clustering with a pre-determined number of clusters was performed for three cluster solution. Thus, the final cluster Centres which describe the mean value of each variable for each of the clusters obtained for three cluster solution and the results are given below.

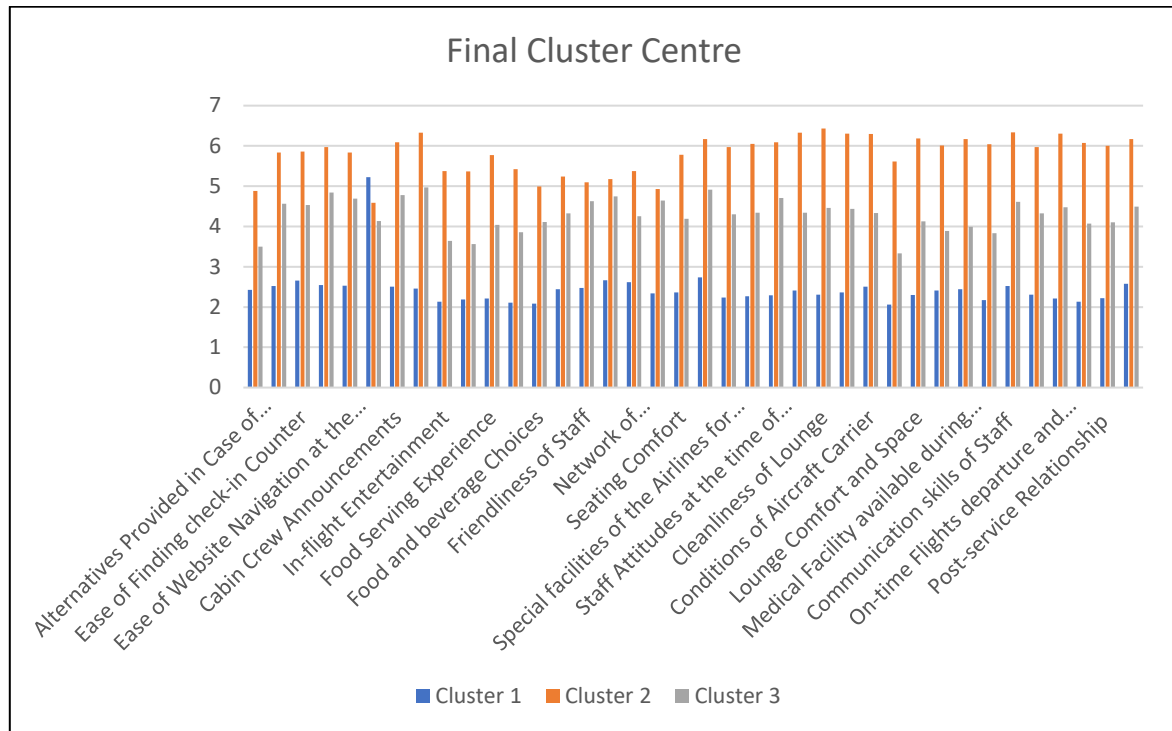


Figure 1: Final Clusters Centre

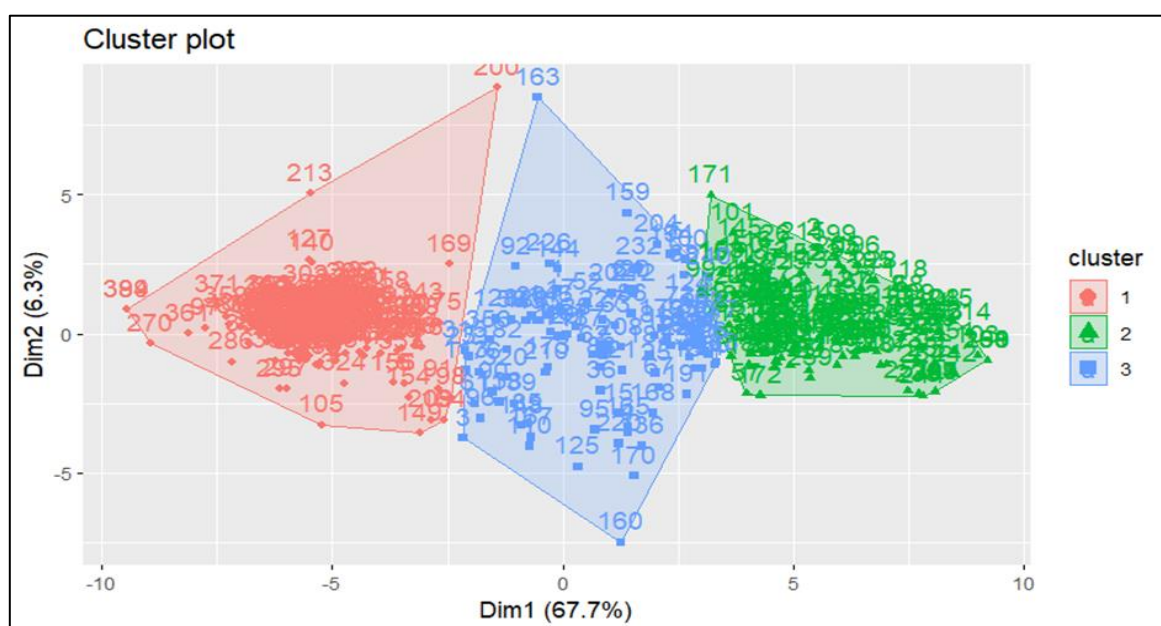


Figure 2: Cluster Plot

ANOVA TABLE						
	Cluster		Error		F	Sig.
	Mean Square	df	Mean Square	df		
Alternatives Provided in Case of Flight Delay	256.752	2	2.27	436	113.109	0.0000
Boarding Announcement and Procedure	476.368	2	1.793	436	265.619	0.0000
Ease of Finding check-in Counter	442.325	2	1.603	436	275.939	0.0000
Ease of Getting Ticket	518.744	2	1.408	436	368.443	0.0000
Ease of Website Navigation at the time of Booking	480.307	2	1.433	436	335.127	0.0000
Waiting times during Check-In	41.168	2	3.311	436	12.434	0.0000
Cabin Crew Announcements	561.104	2	1.353	436	414.725	0.0000
Cleanliness of Cabin	658.937	2	1.141	436	577.368	0.0000
In-flight Entertainment	447.703	2	1.863	436	240.303	0.0000
Leg room available on Seat	429.731	2	1.556	436	276.245	0.0000
Food Serving Experience	539.194	2	1.154	436	467.401	0.0000
Quantity of Food Given	469.774	2	1.293	436	363.225	0.0000
Food and beverage Choices	378.332	2	2.081	436	181.793	0.0000
Efficiency of Check in Staff	347.105	2	2.017	436	172.12	0.0000

Friendliness of Staff	327.732	2	2.043	436	160.435	0.0000
Grooming and Appearance of Staff	301.42	2	1.968	436	153.151	0.0000
Network of destinations/Connectivity	328.641	2	2.103	436	156.305	0.0000
Innovative marketing schemes like special pricing, holiday packages etc.	333.702	2	2.087	436	159.891	0.0000
Seating Comfort	501.257	2	1.299	436	385.985	0.0000
Smooth Take-off/Landing	514.304	2	1.108	436	464.103	0.0000
Special facilities of the Airlines for Children, the Elderly and the Handicapped	599.552	2	1.165	436	514.795	0.0000
Staff Assistance at the time of Arrival	613.604	2	1.11	436	552.891	0.0000
Staff Attitudes at the time of Check-In	630.393	2	1.063	436	593.119	0.0000
Cleanliness of Check-in Counter	652.698	2	0.975	436	669.155	0.0000
Cleanliness of Lounge	726.361	2	0.88	436	825.403	0.0000
Cleanliness of Washrooms	662.352	2	1.004	436	659.723	0.0000
Conditions of Aircraft Carrier	610.213	2	0.795	436	767.814	0.0000
Internet / WIFI / Sky phone options during Flight	542.315	2	1.506	436	360.107	0.0000
Lounge Comfort and Space	642.435	2	0.925	436	694.458	0.0000
Luggage delivery times	554.058	2	1.107	436	500.324	0.0000

Medical Facility available during Flight	590.182	2	0.864	436	682.876	0.0000
Newspaper and Magazines service during Flight	636.364	2	1.173	436	542.299	0.0000
Communication skills of Staff	623.677	2	0.922	436	676.273	0.0000
Dissemination of Information in case of Delay of Flight	575.449	2	1.007	436	571.649	0.0000
On-time Flights departure and arrivals	718.779	2	1.001	436	717.737	0.0000
Personalized attention given to you	663.846	2	0.932	436	712.045	0.0000
Post-service Relationship	608.878	2	0.966	436	630.618	0.0000
Security of the Luggage	550.73	2	0.962	436	572.69	0.0000

Table 16: ANOVA Table of Cluster Analysis

5.3 DISCUSSION

The K-means clustering process chose to maximise the differences among cases in different clusters (Nargundkar, 2007). An inter-cluster comparison (figure 1) shows that the clusters are fairly unique in terms of the customer's preference towards airline services behaviour. Passengers in Cluster 1 generally exhibit low satisfaction across most service attributes. Passengers in Cluster 2 demonstrate moderate to high levels of satisfaction across a wide range of service attributes. They rate significantly higher compared to Cluster 1 on most variables, indicating overall better experiences. Cluster 3 passengers exhibit moderate satisfaction levels across most service attributes, falling between the ratings of Clusters 1 and 2.

Cluster 1 which is the largest cluster consist of 185 individuals (42%) of total 440 customers, exhibits notably low satisfaction scores across a range of airline service attributes, reflecting a general trend of dissatisfaction among passengers within this segment. The ratings for various aspects such as alternatives provided in case of flight delay, boarding announcement and procedure, ease of finding check-in counter, and other service-related factors are considerably low, indicating significant areas of improvement needed in service delivery. Specifically, passengers in Cluster 1 express dissatisfaction with waiting times during check-in and in-flight

entertainment, highlighting specific pain points that airlines should address to enhance overall customer satisfaction and improve the passenger experience. These insights can be used to prioritize service enhancements and targeted improvements tailored to the needs and preferences of passengers in Cluster 1, ultimately aiming to elevate satisfaction levels and foster greater customer loyalty within this segment.

Cluster 2 consists of 168 individuals (38%) of total 440 customers, it exhibits notably higher satisfaction scores across a diverse set of airline service attributes, indicating a trend of moderate to high satisfaction among passengers within this segment. Passengers in Cluster 2 rate significantly higher compared to Cluster 1 and Cluster 3 on most variables, reflecting overall positive experiences with airline services. Notable areas of satisfaction include cleanliness of cabin, smooth take off/landing experiences, and effective communication skills of staff. These positive ratings suggest that Cluster 2 passengers are generally pleased with the quality of service received, highlighting strengths that airlines can leverage to further enhance customer experiences and reinforce brand loyalty. By capitalizing on these positive aspects and maintaining service excellence, airlines can continue to meet and exceed the expectations of passengers in Cluster 2, ultimately driving customer satisfaction and retention within this valuable customer segment.

Cluster 3 consists of 87 individuals (20%) of total 440 customers, it exhibits moderate satisfaction scores across various airline service attributes, positioning passengers within this segment between the lower satisfaction levels of Cluster 1 and the higher satisfaction levels of Cluster 2. While not as dissatisfied as Cluster 1, passengers in Cluster 3 express a level of satisfaction that could benefit from targeted improvements in specific areas. Attributes such as ease of getting tickets, cleanliness of cabin, and communication skills of staff receive moderate ratings from Cluster 3 passengers. Airlines should focus on addressing these moderate satisfaction levels by further refining services and addressing customer feedback to elevate overall satisfaction and meet the evolving preferences of passengers within Cluster 3.

Implementing strategic enhancements based on these insights can help airlines enhance customer experiences and foster stronger relationships with passengers in this segment.

From Anova Table 16, across all service attributes listed in the table, The F-statistic measures the ratio of variability between group means to variability within groups. A higher F-value indicates a larger difference between cluster means relative to the variability within clusters. For all service attributes listed, the p-values are extremely small (Sig. = .000), indicating highly significant differences in mean scores across clusters. This suggests that there are substantial variations in satisfaction levels for each service attribute among the clusters, and these differences are unlikely

to occur by random chance alone. This implies that the identified clusters (Cluster 1, Cluster 2, Cluster 3) exhibit notably different satisfaction levels across a wide range of airline service attributes.

Airlines should adopt a segmented approach based on distinct passenger clusters to address varying levels of satisfaction effectively. Tailored strategies focusing on specific pain points and strengths within each segment can lead to more impactful service enhancements. The analysis identifies key areas requiring immediate attention, such as reducing waiting times during check-in and enhancing in-flight entertainment for passengers in Cluster 1. Prioritizing these improvement areas can significantly elevate overall satisfaction levels. Airlines should leverage strengths highlighted by Cluster 2, such as cabin cleanliness and effective staff communication, to reinforce positive experiences and differentiate themselves in the competitive landscape. Implementing targeted improvements based on customer feedback is crucial for meeting evolving passenger preferences and maintaining high levels of satisfaction across all clusters and by consistently delivering exceptional experiences tailored to specific cluster needs, airlines can strengthen brand loyalty, attract new customers, and gain a competitive edge in the market. In conclusion, the analysis underscores the importance of customer-centric strategies in the airline industry. By understanding and addressing the diverse satisfaction levels and preferences of passenger clusters, airlines can optimize service delivery. Tailored strategies focusing on specific pain points and strengths within each segment can lead to more impactful service enhancements. Airlines should prioritize improvement areas highlighted by the analysis, such as reducing waiting times during check-in and enhancing in-flight entertainment for certain passenger clusters. By consistently delivering exceptional experiences tailored to specific cluster needs, airlines can strengthen brand loyalty, attract new customers, and gain a competitive edge in the market.

CHAPTER VI: PASSENGER LOYALTY INSIGHTS: DECISION TREE ANALYSIS IN THE INDIAN DOMESTIC AVIATION SECTOR

6.1 INTRODUCTION

The Indian domestic aviation sector is characterized by a diverse passenger base with varying needs, preferences, and expectations. As airlines strive to differentiate themselves and deliver exceptional service experiences, it becomes imperative to decipher the underlying factors that influence passenger decisions, satisfaction levels, and loyalty behaviours.

Our objectives in this chapter are twofold: firstly, to explore the myriad factors that shape passenger preferences and satisfaction levels, thereby influencing their likeliness to travel again within the Indian domestic aviation sector; and secondly, to delve into the determinants of passenger loyalty and their inclination to recommend airlines to others. Through the lens of decision tree analysis, we aim to unearth hidden patterns, decision rules, and segmentation insights that illuminate the path to passenger satisfaction and loyalty.

By leveraging a comprehensive set of service quality variables as independent factors and employing likeliness to travel again and likelihood to recommend as dependent variables, we embark on a quest to unravel the intricate web of passenger decision-making. Through rigorous analysis and interpretation of decision tree models, we seek to distil actionable insights that inform strategic decision-making processes within the Indian domestic aviation sector.

As we navigate through the branches of decision trees, we gain a deeper understanding of the factors that drive passenger preferences, satisfaction, and loyalty. By uncovering key drivers and segmentation patterns, we aim to equip airlines with the knowledge and insights needed to optimize service delivery, enhance customer experiences, and foster enduring relationships with their passengers.

6.2 METHODOLOGY

6.2.1 Data Collection

We collected primary data through a structured survey administered to passengers traveling within the Indian domestic aviation sector as well as with the use of Google form. The survey captured a comprehensive set of service quality variables, including aspects related to booking, check-in, onboard experience, staff interactions, cleanliness perceptions, and overall satisfaction ratings.

A total of 440 respondents participated in the survey, providing valuable insights into their preferences, satisfaction levels, and loyalty behaviours.

6.2.2 Software Utilization

In this section, we delineate the software utilized in our study, namely IBM SPSS, to conduct decision tree analysis and subsequent interpretation of results.

6.2.3 Decision Tree Analysis

Software Utilization:

In this study, decision tree analysis was conducted using IBM SPSS. In IBM SPSS, we utilized two distinct methods for decision tree analysis: CHAID (Chi-squared Automatic Interaction Detection) and CRT (Classification and Regression Trees).

Decision Tree Models:

For decision tree analysis in IBM SPSS, we first employed the CHAID method, which is a non-parametric technique for creating tree-like models. The CHAID algorithm recursively partitions the dataset into mutually exclusive and exhaustive segments based on the significance of predictor variables in predicting the target variable. Subsequently, we utilized the CRT method, which constructs decision trees based on statistical significance tests to determine the optimal splits at each node. Both CHAID and CRT methods were used to generate decision tree models for likeliness to travel again and likelihood to recommend in the Indian domestic aviation sector.

Parameter Settings and Evaluation:

In IBM SPSS, decision tree analysis was performed with specific parameter settings tailored to the characteristics of the dataset and research objectives. Parameters such as minimum node size, maximum tree depth, and splitting criteria were adjusted to optimize the performance and interpretability of the decision tree models. Model evaluation was conducted using techniques such as cross-validation and goodness-of-fit tests to assess the reliability and generalizability of the models.

Comparison and Validation:

The decision tree models generated using different methods in IBM SPSS (CHAID, CRT) were compared and evaluated to ensure consistency and validity of the results. By examining the similarities and differences between the models, we aimed to gain a comprehensive understanding of the factors influencing passenger preferences, satisfaction levels, and loyalty behaviours within the Indian domestic aviation sector.

6.2.4 Interpretation of the result

Various approaches and possibilities are tried and the resulting decision tree models were analysed to identify important predictors and decision rules that influence passenger preferences, satisfaction levels, and loyalty behaviours.

Segmentation patterns and distinct groups of passengers were identified based on their characteristics and behaviours, providing insights into the heterogeneity of passenger preferences and loyalty dynamics within the Indian domestic aviation sector.

6.2.5 Limitations

While decision tree analysis provides valuable insights into passenger preferences and loyalty dynamics, it is not without limitations.

Like any other method, Decision tree analysis comes with its own set of limitations. The accuracy and interpretability of decision tree models may be affected by factors such as sample size, variable selection, and the complexity of interactions between variables.

Additionally, decision tree analysis assumes independence between variables at each decision point, which may not always hold true in real-world scenarios.

6.2.6 Conclusion

Despite these limitations, decision tree analysis serves as a powerful tool for uncovering the factors influencing passenger preferences, satisfaction levels, and loyalty behaviours within the Indian domestic aviation sector.

By leveraging the insights gained from decision tree analysis, airlines can tailor their offerings and strategies to better meet the needs and expectations of their passengers, thereby fostering greater satisfaction, loyalty, and advocacy.

Thus, using the decision tree analysis, we paved the pathway of Continuous improvement and loyalty of passengers for Indian domestic aviation sector for a bright future.

6.3 ANALYSIS

In this section, we outline the methodology employed to explore passenger preferences, satisfaction levels, and loyalty dynamics within the Indian domestic aviation sector using decision tree analysis. Our approach is guided by the objectives of uncovering the factors influencing passenger preferences and satisfaction levels, as well as understanding the determinants of passenger loyalty and their likeliness to recommend airlines to others.

The "Analysis" section serves as the nucleus of our study, delving deep into the intricacies of passenger preferences, satisfaction levels, and loyalty dynamics within the Indian domestic aviation sector. In this section, we embark on a comprehensive exploration of the insights gleaned from our multifaceted analysis approaches, each tailored to provide a nuanced understanding of the factors driving passenger behaviour and perceptions.

With the aim of uncovering the underlying determinants of passenger preferences and loyalty, we employed three distinct approaches, each offering unique perspectives and insights into the complex interplay of variables shaping passenger decision-making. Firstly, we conducted a traditional analysis, employing standard statistical techniques to examine the relationships between various demographic and service quality variables and passengers' likeliness to travel again and recommend airlines to others.

In parallel, we also explored the correlation-based variable selection approach, leveraging the relationships between variables to identify key predictors of passenger preferences and loyalty. Through rigorous statistical analysis and data-driven decision-making, we curated a subset of variables with the highest positive correlations with our dependent variables, thus focusing our attention on the most influential factors driving passenger satisfaction and loyalty.

Furthermore, we harnessed the analytical prowess of a powerful and user-friendly software tools—IBM SPSS—to conduct decision tree analysis, a robust and versatile method for uncovering intricate patterns and decision rules within categorical data. By employing decision tree analysis techniques such as CHAID and CRT, we aimed to segment passengers based on their characteristics and behaviours, thus gaining deeper insights into the heterogeneity of passenger preferences and loyalty dynamics.

Through this multifaceted analysis journey, we aim to shed light on the nuanced nuances of passenger behaviour and perceptions within the Indian domestic aviation sector, offering actionable insights and recommendations for airlines seeking to enhance their service offerings, optimize customer experiences, and foster lasting relationships with their passengers. Join us as we unravel the complexities of passenger preferences and loyalty, guided by the illuminating insights derived from our rigorous analysis approaches.

6.3.1 Approach 1: Normal Approach

Data Preparation:

- The data is first downloaded as a excel file, this excel file is then imported into IBM SPSS software where it was first cleaned, recoded and transformed.
- In this approach, we have taken a comprehensive set of 38 service quality variables as our independent variables in our decision tree analysis.
- We have use CHAID [Chi-Square automatic interaction detection] as our growing method in our analysis.

Decision Tree Analysis - SPSS:

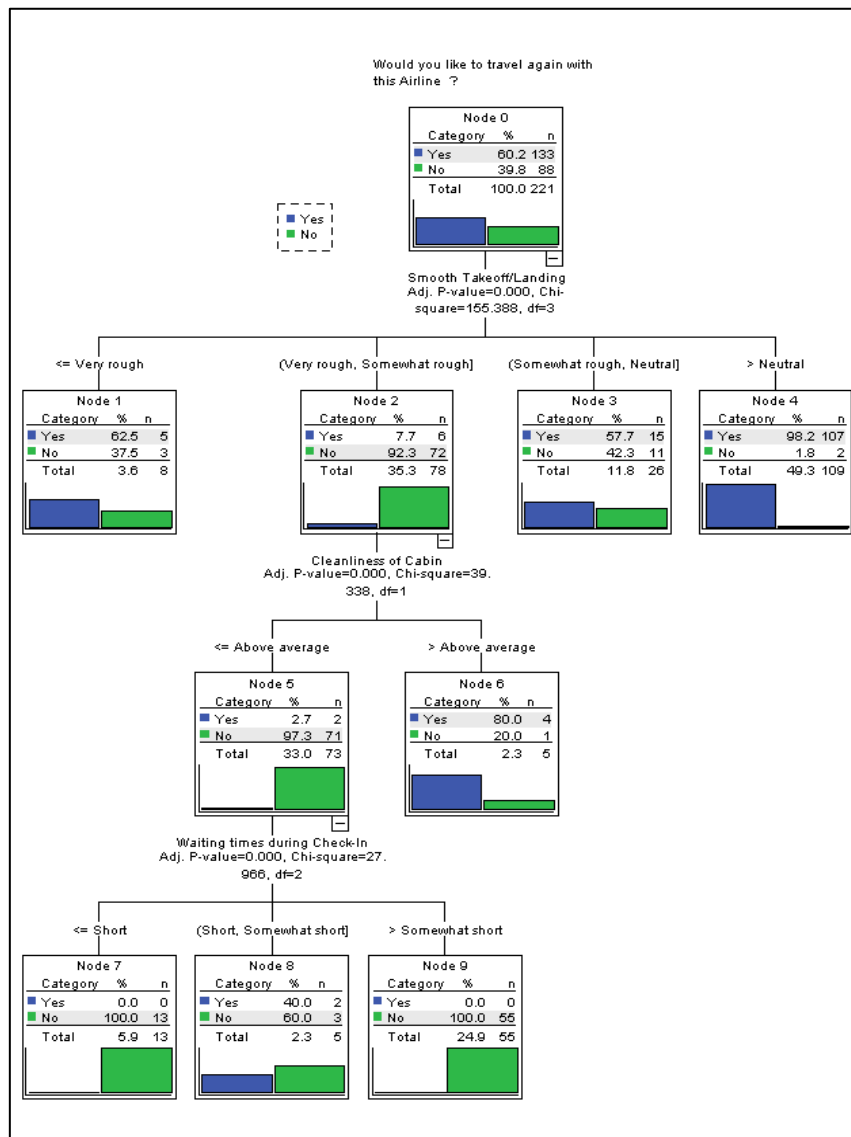


Figure 3: Decision tree for "Would you like to travel again?" [Training set]

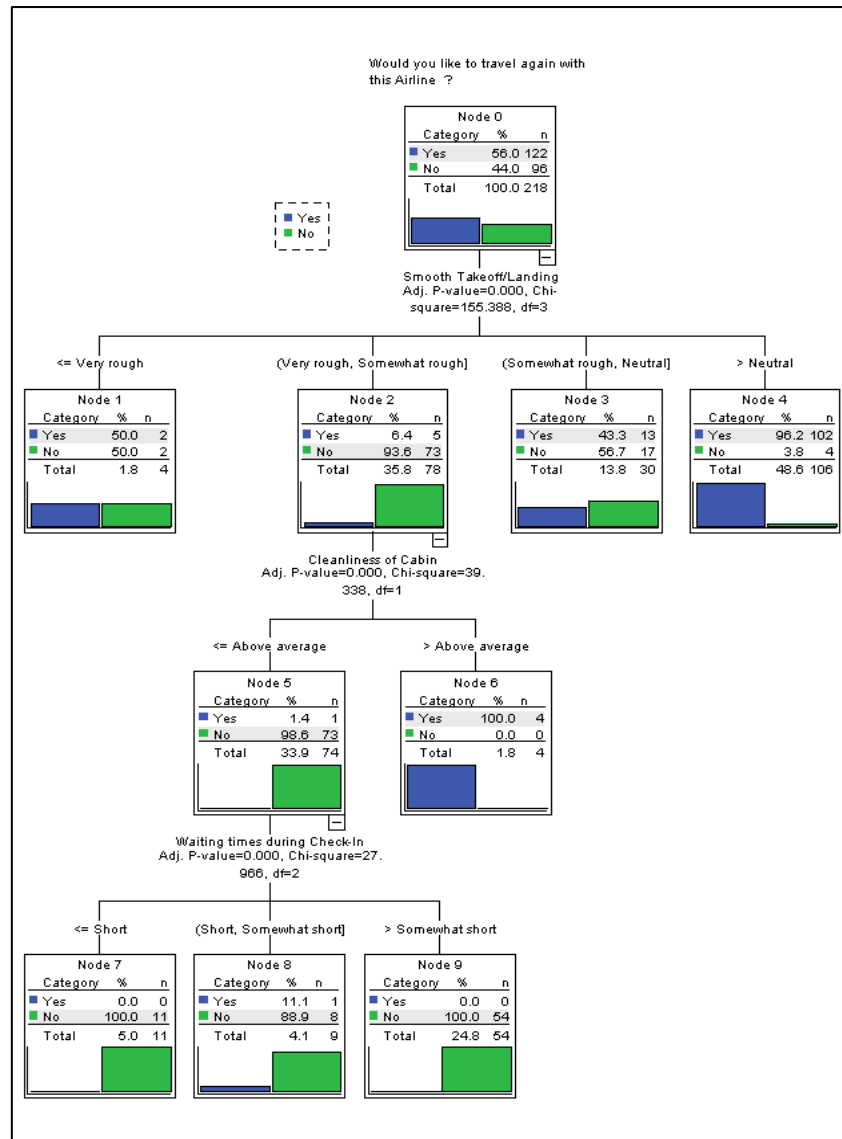


Figure 4: Decision tree for "Would you like to travel again?" [Test set]

Here, we have taken our dataset of Indian Domestic Aviation sector for 440 respondents. The data is further split into training set and test set and the Decision tree analysis was performed. The goal of the analysis was to identify the most prominent service quality variables from a pool of 38 service quality variables that will influence passenger's likeliness to travel again by the Indian Domestic Airlines.

Here, the results of our decision tree analysis are illustrated Figure 3 and Figure 4 for training and testing sets respectively.

Our model achieves the accuracy of 91.4 % on the training set, indicating that it correctly predicts likeliness to travel again with the airline 91.4 % of time. Similarly, the model achieves a similar accuracy of 89 % on the test set, indicating good generalization ability. This means it performs well not only on the data it was trained on but also on new, unseen data.

So, by comparing the performance we can say that our model is not overfitting to the trained data and can make reliable predictions on new data.

Based on the Decision tree diagram shown above, we can see that, “Smooth Take-off/Landing” is the most important factor, followed by “Cleanliness of cabin” in customers likeliness to travel again by airlines.

We can see that, customers who give review value as greater than Neutral for “Smooth Take-off/Landing” are most likely to travel again by Indian Domestic airlines. While customers who give review value as between Very rough to Somewhat rough for “Smooth Take-off/Landing” are least likely to travel again by Indian Domestic airlines.

Also, customers who give review value as less than equal to Above average for “Cleanliness of Cabin” are least likely to travel again by Indian Domestic airlines.

So, based on the tree results we can conclude that, Indian Airlines should pay precise attention and maintain Smoothness consistently during flight’s Take-off and Landing. And along with that also improve and maintain high standards of cleanliness through the cabin.

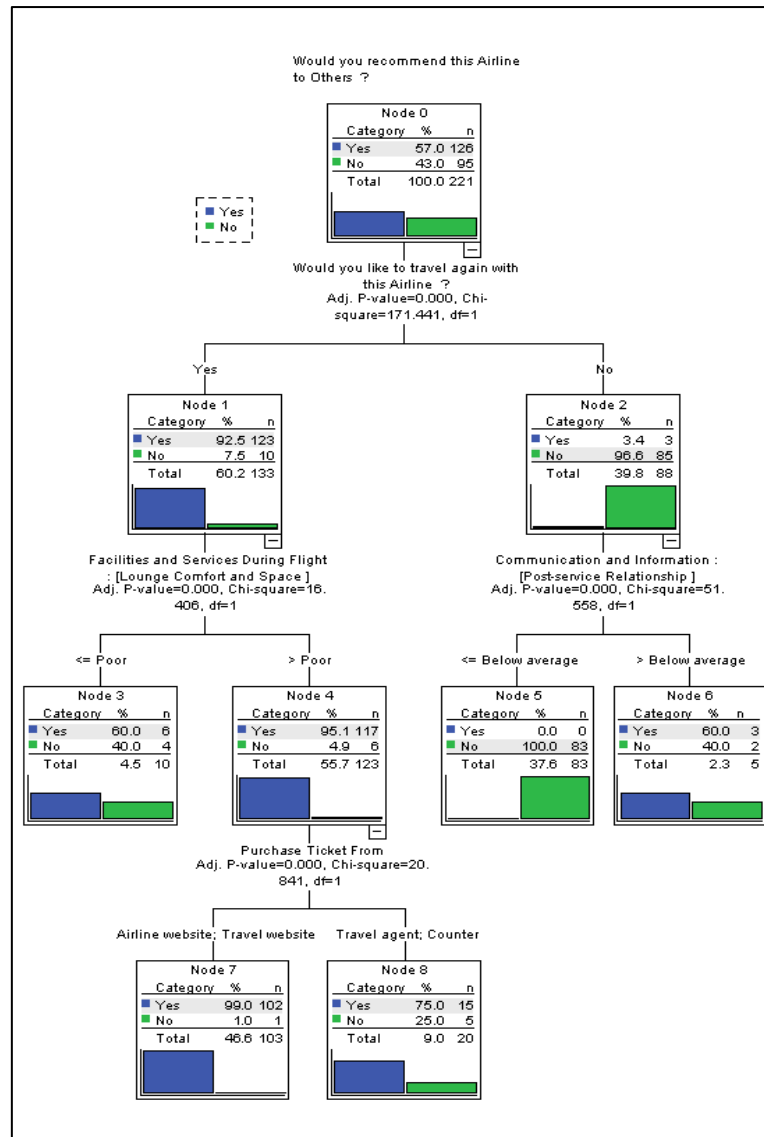


Figure 5: Decision tree for “Would you Recommend this Airline to others?” [Training set]

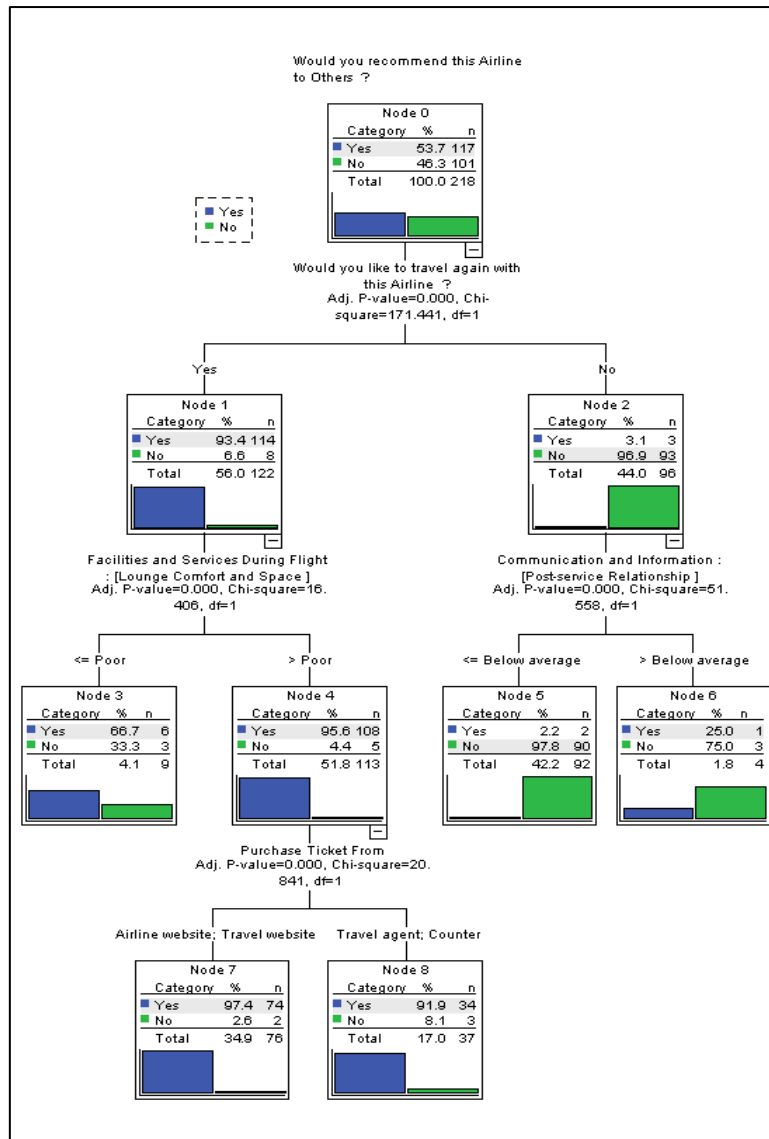


Figure 6: Decision tree for "Would you Recommend this Airline to others?" [Test set]

Here, we have taken our dataset of Indian Domestic Aviation sector for 440 respondents. The data is further split into training set and test set and the Decision tree analysis was performed. The goal of the analysis was to identify the most prominent service quality variables from a pool of 38 service quality variables that will influence passenger's likeliness to Recommend the Indian Domestic Airlines to others.

Here, the results of our decision tree analysis are illustrated and visualized in Figure 5 and Figure 6 for training and testing sets respectively.

Our model achieves the accuracy of 94.6 % on the training set, indicating that it correctly predicts likeliness to travel again with the airline 94.6 % of time. Similarly, the model achieves a similar accuracy of 94 % on the test set, indicating good generalization ability. This means it performs well not only on the data it was trained on but also on new, unseen data.

So, by comparing the performance we can say that our model is not overfitting to the trained data and can make reliable predictions on new data.

Based on the Decision tree diagram shown above, we can see that, “Would you like to travel again with this Airline?” is the most important factor, followed by “Lounge Comfort and space” in customers likeliness to Recommend this airline to others.

We can see that, customers who give review value as Yes for “Would you like to travel again with this Airline?” are most likely to recommend Indian Domestic airlines to others. While customers who give review value as greater than Poor for “Lounge Comfort and space” are second most likely to recommend others.

Also, we can see that customers who give review value as less than equal to Below average for “Post service relationship” are least likely to recommend Indian Domestic airlines to others.

So, based on the tree results we can conclude that, Indian Airlines should pay precise attention on Lounge comfort and space of the airplanes. And along with that should also invest in staff training programs to improve friendliness and post service relationship with the customers.

6.3.2 Approach 2: Correlation-Based variable selection approach

Data Preparation:

- In this approach similar to previous approach the data is downloaded as a excel file from the questionnaire.
- It was later imported into IBM SPSS, where it was cleaned, organized, recoded and transformed.
- Now, in this Approach, unlike previous approach, using SPSS we will first check Spearman’s Rho correlations between our dependent variables of “would you like to travel again?” and “would you recommend this airline to others?” and all the remaining 54 variables.
- We would select only those variables that have positive correlation with our dependent variables.
- From our analysis we got our significant independent variables as: Age, highest Qualification, Purpose of travel, choice decided by, waiting time during check-in, class of travel.
- We are using both CHAID [Chi-Square automatic interaction detection] as well as CRT [Classification and Regression trees] as our growing methods in our analysis.

Decision Tree Analysis - SPSS:

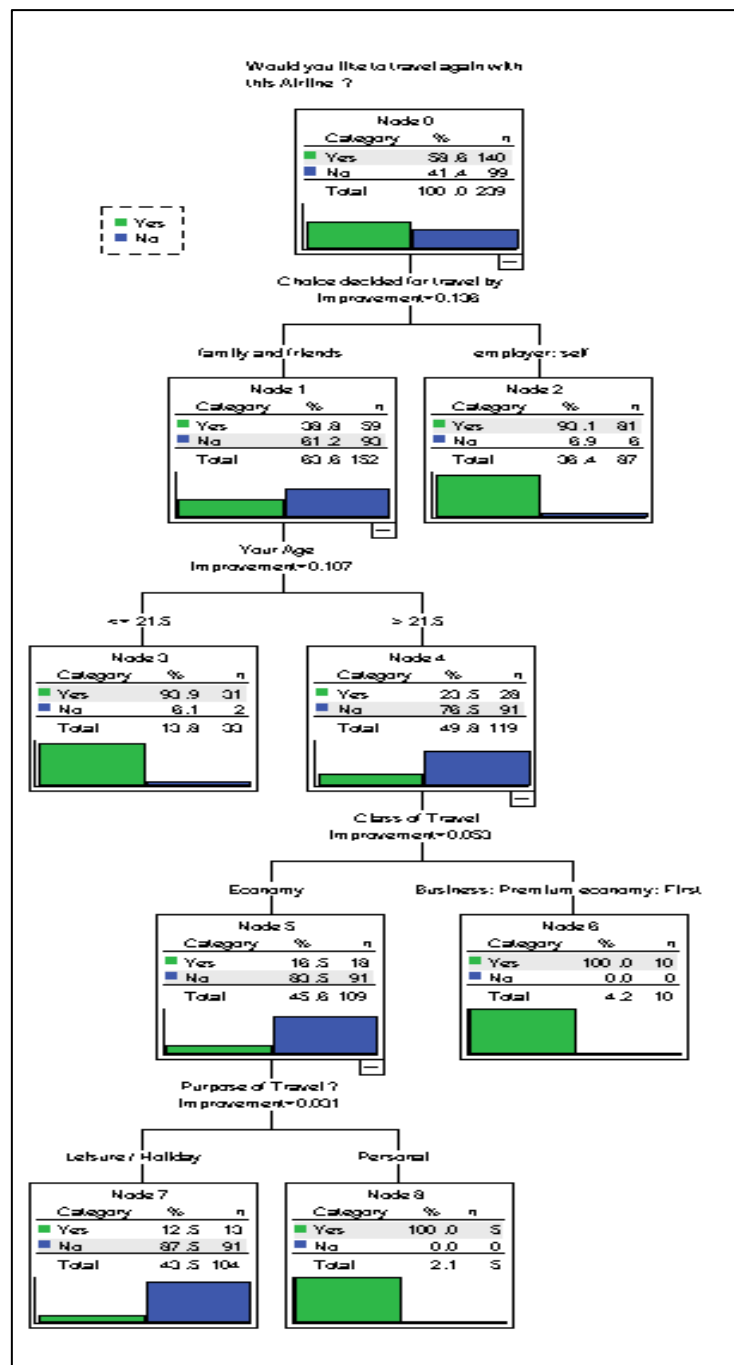


Figure 7: Decision tree for "Would you like to travel again?" [Train set]

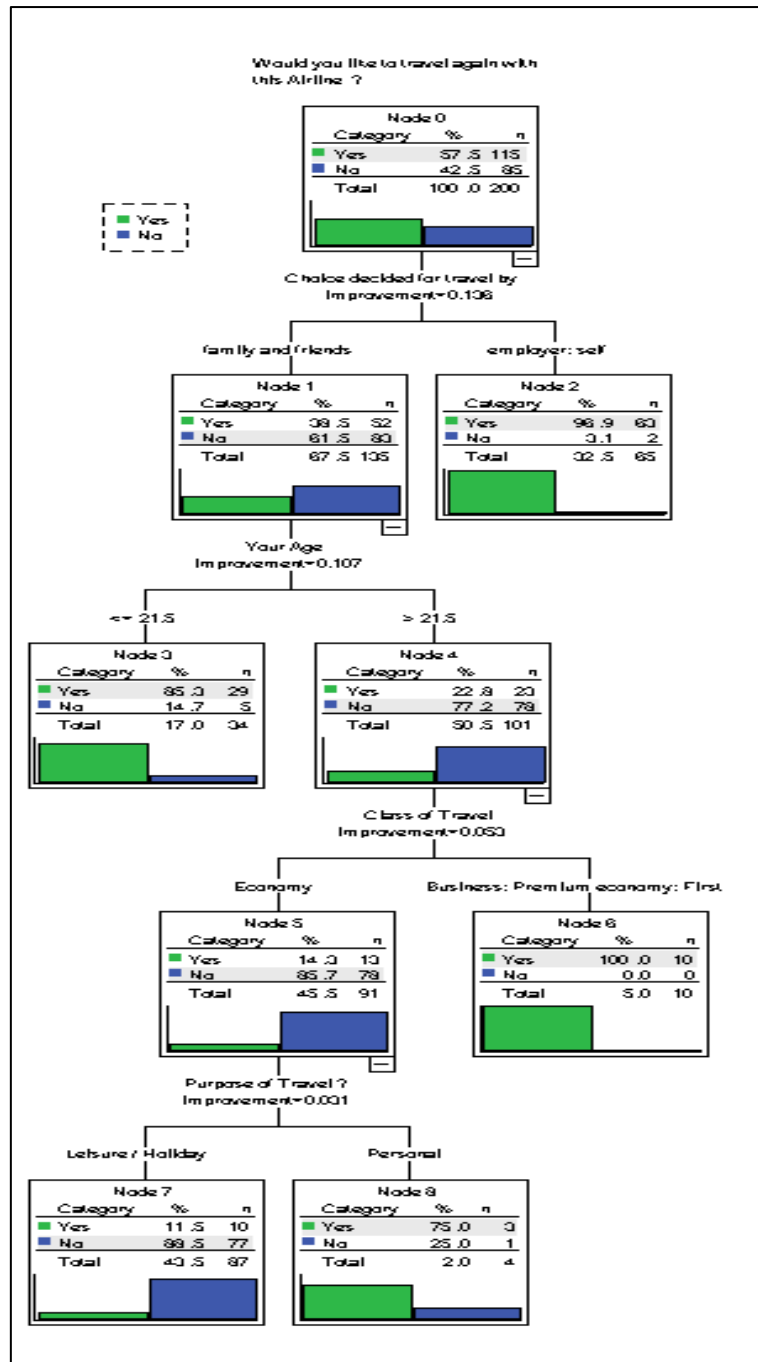


Figure 8: Decision tree for "Would you like to travel again?" [Test set]

Here, we have taken our dataset of Indian Domestic Aviation sector for 440 respondents. The variables which correlate positively with the dependent variables are taken as independent variables. The data is further split into training set and test set and the Decision tree analysis was performed. The goal of the analysis was to identify the most prominent variables that will influence passenger's likeliness to travel again by the Indian Domestic Airlines.

Here, the results of our decision tree analysis are illustrated and visualized in Figure 7 and Figure 8 for training and testing sets respectively.

Our model achieves the accuracy of 91.2 % on the training set, indicating that it correctly predicts likeliness to travel again with the airline 91.2 % of time. Similarly, the model achieves a similar accuracy of 90 % on the test set, indicating good generalization ability. This means it performs well not only on the data it was trained on but also on new, unseen data.

So, by comparing the performance we can say that our model is not overfitting to the trained data and can make reliable predictions on new data.

Based on the Decision tree diagram shown above, we can see that, “Choice decided for travel by” is the most important factor, followed by “Age” in customers likeliness to travel again by airlines.

We can see that, customers whose choice of travel were decided by an Employer or Self are most likely to travel again by Indian Domestic airlines. While customers who are in the age group above 21 years are least likely to travel again by Indian Domestic airlines.

So, based on the tree results we can conclude that, the chances of customers whose choice of travel was planned by Self or an Employer are most likely to travel again by Indian Domestic Airlines.

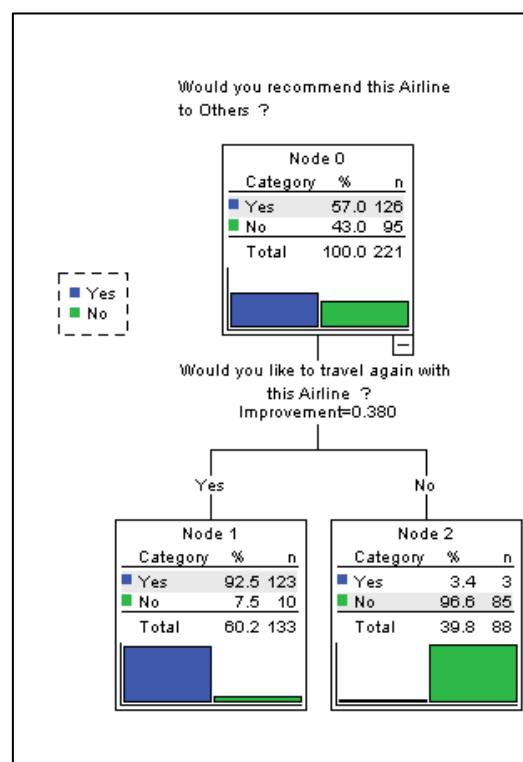


Figure 9: Decision tree for “Would you Recommend this Airline to others?” [Training set]

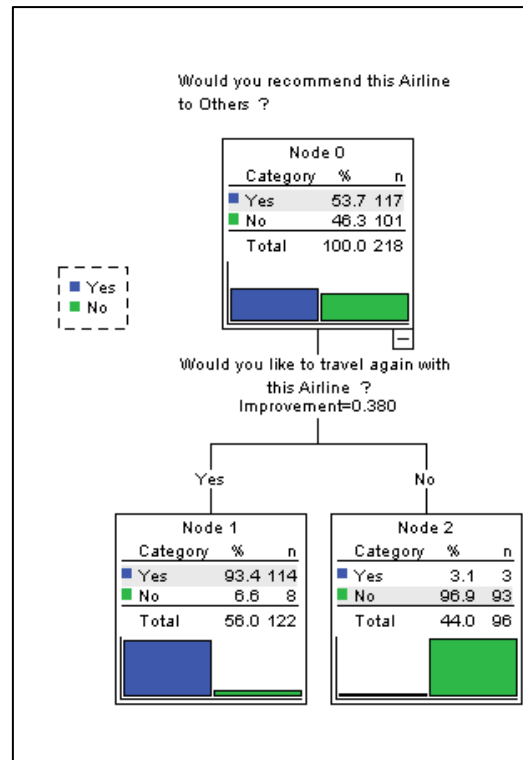


Figure 10: Decision tree for “Would you Recommend this Airline to others?” [Test set]

Here, we have taken our dataset of Indian Domestic Aviation sector for 440 respondents. The variables which correlate positively with the dependent variables are taken as independent variables. The data is further split into training set and test set and the Decision tree analysis was performed. The goal of the analysis was to identify the most prominent variables that will influence passenger’s likeliness to Recommend the Indian Domestic Airlines to others.

Here, the results of our decision tree analysis are illustrated and visualized in Figure 9 and Figure 10 for training and testing sets respectively.

Our model achieves the accuracy of 94.1 % on the training set, indicating that it correctly predicts likeliness to travel again with the airline 94.1 % of time. Similarly, the model achieves a similar accuracy of 95 % on the test set, indicating good generalization ability. This means it performs well not only on the data it was trained on but also more well on new, unseen data.

So, by comparing the performance we can say that our model is not overfitting to the trained data and can make reliable predictions on new data.

Based on the Decision tree diagram shown above, we can see that, “Would you like to travel again with this Airline?” is the most important factor, in customers likeliness to Recommend this airline to others.

We can see that, customers who give review value as Yes for “Would you like to travel again with this Airline?” are most likely to recommend Indian Domestic airlines to others. While customers

who give review value as No for “Would you like to travel again with this Airline?” are least likely to recommend Indian Domestic airlines to others.

So, based on the tree results we can conclude that, Indian Airlines should pay precise attention on improving and maintaining factors that satisfies customers and usurp their loyalty.

6.3.3 Conclusion

So, by thoroughly going through both the approaches we got to know that:

- Whether a customer or a passenger would recommend the Airline to others will mostly depend on how was their overall experience and whether they themselves like to travel again by that airlines or not.
- Indian Domestic aviation sector should work on major domains affecting customer’s likeliness to travel again with the Airlines.
- They should pay precise attention and maintain Smoothness consistently during flight’s Take-off and Landing. And along with that also improve and maintain high standards of cleanliness through the cabin.
- They should pay precise attention on Lounge comfort and space of the airplanes. And along with that should also invest in staff training programs to improve friendliness and post service relationship with the customers.

CHAPTER VII: PREDICTORS OF PASSENGER LOYALTY: LOGISTIC REGRESSION ANALYSIS

7.1 INTRODUCTION

In the fiercely competitive landscape of the airline industry, understanding passenger behaviour and factors influencing their decision to travel again is of paramount importance for airlines striving to maintain a loyal customer base and sustainable profitability. Passenger experience, shaped by a multitude of socio-economic and demographic factors, plays a pivotal role in determining satisfaction, loyalty, and ultimately, the success of airlines.

This chapter presents an in-depth analysis of passenger experience in Indian airlines using binary logistic regression. By investigating the probability of passengers choosing to travel again and the effects of socio-economic and demographic factors on this probability, this study aims to shed light on the intricacies of passenger decision-making in the Indian aviation context.

The primary objective of this study is to analyse the probability of passengers choosing to travel again with Indian airlines i.e. to investigate the influence of socio-economic factors (e.g., income, education level) and demographic characteristics (e.g., age, gender) on passenger choice to travel again.

In binary logistic regression, the outcome variable is dichotomous, meaning it has only two possible outcomes. In this case, the outcome variable would indicate whether a passenger chooses to travel again (1) or not (0). The predictor variables, on the other hand, can be continuous (e.g., income) or categorical (e.g., gender, education level). In the context of analysing passenger behaviour in Indian airlines, binary logistic regression allows researchers to identify which socio-economic and demographic factors significantly influence the likelihood of passengers choosing to travel again. By understanding these factors, airlines can tailor their marketing strategies, service offerings, and customer experience initiatives to enhance passenger satisfaction and loyalty.

7.2 METHODOLOGY

7.2.1 Data Collection

We collected primary data through a structured survey administered to passengers traveling within the Indian domestic aviation sector as well as with the use of Google form. A total of 440 respondents participated in the survey, providing valuable insights into their preferences, satisfaction levels, and loyalty behaviours.

7.2.2 Software Utilization

In this section, we delineate the software utilized in our study, namely IBM SPSS, to conduct binary logistic regression and subsequent interpretation of results.

7.2.3 Variable Selection

We identified a set of independent variables representing various dimensions of socio-economic, demographics and travel behaviour. These variables were selected based on their relevance to the objectives of the study and their potential to influence passenger's choice to travel again.

Likeness to travel again was chosen as the dependent variables while Gender, Profession, Monthly Income, Residential Status, Highest Qualification, Purpose of Travel, Frequency of Travel, Class of Travel, Choice decided for travel by & Purchase Ticket From are chosen as independent variables.

7.3 ANALYSIS

Iteration		-2 Log likelihood	Coefficients
			Constant
Step 0	1	598.135	-.327
	2	598.135	-.330
	3	598.135	-.330

Table 17: Iteration History at Step 0

Observed			Predicted		
			Would you like to travel again with this Airline?		Percentage Correct
			Yes	No	
Step 0	Would you like to travel again with this Airline?	Yes	256	0	100.0
		No	184	0	.0
	Overall Percentage				58.2

Table 178: Classification Table at Step 0

From Table 18, the overall percentage of observations that were correctly classified is 58.2%

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-.330	.097	11.675	1	.001	.719

Table 18: Variables in the Equation

Table 19 clearly states that the constant is -0.33 and it is highly significant with an odds of 0.719

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	447.494 ^a	.290	.390
2	383.281 ^a	.386	.520
3	335.238 ^a	.450	.605
4	308.938 ^b	.482	.648
5	293.500 ^b	.500	.672
6	279.684 ^b	.515	.693
7	273.315 ^b	.522	.702
8	263.023 ^b	.533	.717

Table 19: Model Summary

Table 20 gives the -2 Log likelihood values and Nagelkerke R square values also known as pseudo R square. With every new step the -2 Log likelihood value decreases while pseudo R square value increases and it stopped at step 8. This indicates that it took 8 steps to reach the correct model where 71.7% variation in dependent variable is explained by all the independent variables.

Step	Chi-square	df	Sig.
1	.000	1	1.000
2	13.082	4	.011
3	30.398	5	.000
4	54.633	5	.000
5	84.535	5	.000
6	229.801	6	.000
7	257.057	8	.000
8	143.507	8	.000

Table 20: Hosmer and Lemeshow Test

Hypothesis:

H_0 : There is no difference between the observed and model-predicted values

H_1 : There is difference between the observed and model-predicted values

From table 21 of Hosmer and Lemeshow Goodness of Fit test, it is found that every step is significant. It indicates that our model fit is good.

Observed			Predicted		
			Would you like to travel again with this Airline?		Percentage Correct
			Yes	No	
Step 8	Would you like to travel again with this Airline?	Yes	231	25	90.2
		No	17	167	90.8
	Overall Percentage				90.5

Table 21: Classification Table at Step 8

From Table 22, at the final step the overall percentage of observations that were correctly classified is 90.5%

Background Characteristics	Estimate (B)	Sig	Odds Ratio (Exp(B))	Lower CI for Odds Ratio	Upper CI for Odds Ratio
Gender					
Female ®					
Male	-.667	.048	.513	.265	.995
Profession					
Others ®					
Public	-1.204	.057	.300	.087	1.038
Private	-1.187	.073	.305	.083	1.120
Government	-.793	.281	.452	.107	1.914
Business	-1.782	.004	.168	.050	.565
Monthly Income					
More than Rs. 3 Lakh ®					

Less than Rs. 50,000	-.525	.304	.591	.217	1.610
Rs. 50,000 – 1 Lakh	-.007	.991	.993	.317	3.112
Rs. 1-2 Lakh	-.694	.364	.499	.111	2.239
Rs. 2-3 Lakh	-3.522	.002	.030	.003	.281
Residential Status					
Other ®					
Own	-2.171	.001	.114	.032	.413
Rent	-.607	.495	.545	.095	3.114
Purpose of Travel					
Leisure/ Holiday ®					
Business	-1.933	.088	.145	.016	1.337
Personal	1.076	.257	2.932	.456	18.859
Frequency of Travel					
Very Frequently ®					
Rarely	-1.698	.000	.183	.074	.454
Occasionally	-.701	.414	.496	.092	2.668
Frequently	-1.627	.144	.197	.022	1.746
Class					
First ®					
Economy	-3.106	.005	.045	.005	.385
Premium Economy	-.479	.700	.619	.054	7.113
Business	-1.532	.079	.216	.039	1.196
Choice decided for travel by					
Employer ®					
Self	3.006	.000	20.199	7.294	55.937
Friends/ Family	-.183	.898	.833	.051	13.620
Constant		.532	.528		

Table 22: Effect of socio-economic and demographic characteristics on passenger choice to travel again

7.4 CONCLUSION

The table 23 shows the odds ratio, regression estimates, significance level and 95 percent confidence interval for the effect of various socio-economic and demographic factors on passengers' choice to travel again. The odds ratio for each characteristic represents the likelihood

of travelling again compared to a reference group. The odds of travelling again were lower for male with odds ratio of 0.513 as compared to women. The odds of travelling again among passengers who travelled for personal reasons was 2.93 which was three times higher than the odds of passengers who travelled for leisure or holidays indicating a very strong relationship between purpose of travel and choice to travel again.

The odds of travelling again for passengers travelling rarely, occasionally and frequently were lower with odds of 0.183, 0.496 & 0.197 respectively as compared to very frequently travelling passengers.

Hence, the logistic model obtained is:

$$\ln \left(\frac{p}{1-p} \right) = -0.33 - 0.667[\text{MALE}] - 1.204[\text{PUBLIC}] - 1.782[\text{BUSINESS}] \\ - 3.522[\text{RS. 2-3 LAKH}] - 2.171[\text{OWN}] - 1.698[\text{RARELY}] - 3.106[\text{ECONOMY}] + 3.006[\text{SELF}]$$

So, odds ratio is,

$$\text{OR} = \exp (-0.33 - 0.667[\text{MALE}] - 1.204[\text{PUBLIC}] - 1.782[\text{BUSINESS}] \\ - 3.522[\text{RS. 2-3 LAKH}] - 2.171[\text{OWN}] - 1.698[\text{RARELY}] - 3.106[\text{ECONOMY}] + 3.006[\text{SELF}])$$

7.5 DISCUSSION

Our analysis identified several significant predictors of passengers choosing to travel again, including age, income, education, and gender. These findings underscore the importance of considering a diverse range of factors in understanding passenger behaviour.

Notably, not all levels of categorical predictors were found to be significant, highlighting the heterogeneity in the effects of socio-economic and demographic variables on passenger decisions. This underscores the need for a nuanced understanding of passenger preferences and motivations.

In summary, this study contributes to the growing body of literature on passenger experience in Indian airlines by employing binary logistic regression to analyse the probability of passengers choosing to travel again as well as highlights the significance of Socio-economic and Demographic factors in shaping passenger behaviour.

Airlines should consider these factors when designing marketing strategies and service offerings. Strategies aimed at enhancing passenger loyalty should focus on addressing the significant predictors identified in the analysis, such as age, income, education, and gender. By understanding the predictors of passenger loyalty, airlines can tailor their services to meet the

needs and preferences of different passenger segments, thereby driving customer satisfaction and loyalty in the Indian aviation sector.

By elucidating the socio-economic and demographic factors shaping passenger behaviour, this research aims to inform evidence-based decision-making in the airline industry and pave the way for enhanced customer satisfaction and loyalty in Indian aviation.

CHAPTER VIII: PASSENGER SENTIMENT ANALYSIS: EVALUATING CUSTOMER FEEDBACK IN THE INDIAN DOMESTIC AVIATION SECTOR

8.1 METHODOLOGY

To understand the overall emotion of the passengers travelling in Indian Airlines, the study questionnaire included a question “Any additional comments or suggestions”. This was an open-end question where respondents were free to share their travel experience.

This variable was used to carry out sentiment analysis. Sentiment analysis was carried out on the website “DanielSoper.com”. All the comments obtained in this question were pasted in the dialogue box and analysed.

8.2 RESULTS

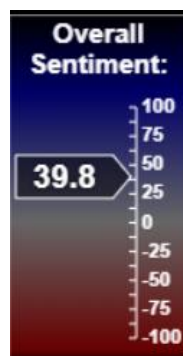


Figure 11: Sentiment Analysis Result

8.3 DISCUSSION

Sentiment scores range from -100 to +100, where -100 indicates a very negative or serious tone and +100 indicates a very positive or enthusiastic tone. Here, the additional comments have a sentiment score of 39.8. This means that the overall sentiment or tone is somewhat positive / enthusiastic. This indicates that the passengers are somewhat happy and positive with the services provided by Indian Airlines.

In conclusion, the overall positive sentiment expressed by passengers in their comments and suggestions is indicative of satisfactory experiences with Indian domestic airlines. However, there is always room for improvement. Airlines should continue to monitor passenger feedback and address any areas of concern or dissatisfaction highlighted by sentiment analysis. By consistently striving to meet passenger expectations and deliver exceptional service, airlines can foster long-term customer loyalty and enhance their reputation in the Indian aviation sector.

CHAPTER IX: RESULTS AND DISCUSSIONS:

UNDERSTANDING PASSENGER SATISFACTION AND LOYALTY

This study delved into the statistical measurement of service quality of Indian domestic airlines from a customer perspective, employing various analytical methods to uncover valuable insights. Here, we present a synthesis of the key findings and discussions from the preceding chapters to offer a comprehensive overview of the study's outcomes.

Loglinear Analysis (Chapter 4): The loglinear analysis revealed significant main effects and interaction effects, providing valuable insights into the demographic influences on customer loyalty within the Indian domestic aviation sector. Understanding these factors allows airlines to tailor their services and promotions to better meet the needs and preferences of different customer segments.

Cluster Analysis (Chapter 5): The cluster analysis identified distinct passenger segments based on their satisfaction levels and preferences towards airline services. By understanding the unique characteristics and needs of each cluster, airlines can implement targeted strategies to enhance customer satisfaction and loyalty within each segment.

Decision Tree Analysis (Chapter 6): The decision tree analysis highlighted key factors influencing passengers' likelihood of recommending the airline to others, emphasizing the importance of factors such as overall experience, flight smoothness, cleanliness, lounge comfort, and staff friendliness. Addressing these factors can significantly impact passenger perceptions and loyalty.

Logistic Regression (Chapter 7): The logistic regression analysis identified significant predictors of passengers choosing to travel again, including age, income, education, and gender. Airlines should consider these factors when designing marketing strategies and service offerings to effectively meet the needs and preferences of different passenger segments.

Sentiment Analysis (Chapter 8): The sentiment analysis of passenger comments revealed an overall positive sentiment towards Indian domestic airlines, indicating satisfactory experiences with the services provided. However, there is always room for improvement, and airlines should

continue to monitor feedback and address any areas of concern to enhance customer satisfaction and loyalty.

Overall Insights:

Collectively, the findings from each chapter underscore the importance of a customer-centric approach in the Indian domestic aviation sector. By understanding the demographic influences, satisfaction levels, predictors of loyalty, and sentiment of passengers, airlines can tailor their services and strategies to meet the diverse needs and preferences of their customers. This customer-centric approach is essential for enhancing satisfaction, fostering loyalty, and maintaining a competitive edge in the dynamic aviation market.

Recommendations:

Based on the findings, we recommend that airlines focus on:

- Tailoring marketing strategies and service offerings to different customer segments.
- Addressing key factors influencing passenger satisfaction and loyalty, such as flight smoothness, cleanliness, and staff friendliness.
- Monitoring passenger feedback continuously and implementing timely improvements to enhance overall service quality.
- Investing in staff training programs to ensure consistent delivery of exceptional customer experiences.
- Leveraging data analytics to gain insights into passenger behavior and preferences, enabling more informed decision-making.

In conclusion, this study provides actionable insights for Indian domestic airlines to enhance service quality, improve customer satisfaction, and foster long-term loyalty. By adopting a customer-centric approach and implementing targeted strategies based on the findings, airlines can differentiate themselves in the competitive market landscape and emerge as leaders in the Indian aviation sector.

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APPENDIX C: LIST OF QUESTIONS

GROUP NAME	QUESTIONS	1	2	3	4	5	6	7
Socio-economic & Demographic variables	1.Gender	Male	Female	Other				
	2.Profession	Public	Private	Government	Business	Others		
	3.Monthly Income	< Rs. 50,000	Rs. 50,000- 1 Lakh	Rs. 1-2 Lakh	Rs. 2-3 Lakh	> Rs. 3 Lakh		
	4.Residential Status	Own	Rent	Others				
	5.Highest Qualification	Secondary	Higher Secondary	Bachelor's Degree	Master's Degree	Doctorate	Professional	Others
	6.Purpose of Travel?	Business	Personal	Leisure				
	7.Frequency of Travel	Rarely	Occasionally	Frequently	Very frequently			
	8.Class of Travel	Economy	Premium Economy	Business	First			
	9.Choice decided for travel by	Self	Family & Friends	Employer				
	10.Purchase Ticket from	Airline Website	Travel Website	Counter	Travel Agent			

	11. Airline used most Frequently	Air Asia India	Air India	Go First	IndiGo	SpiceJet	Vistar a	Others
	12. Alternatives Provided in Case of Flight delay 13. Boarding Announcement and Procedure 14. Cabin Crew Announcements	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Agree	Strongly Agree
	15. Ease of Finding Check-in counter 16. Ease of getting Ticket 17. Ease of Website Navigation at the time of Booking	Very Difficult	Difficult	Somewhat Difficult	Neutral	Somewhat Easy	Easy	Very Easy
	18. Waiting times during Check-in	Very Short	Short	Somewhat Short	Neutral	Somewhat Long	Long	Very Long
	19. In-flight Entertainment	Highly Dissatisfied	Dissatisfied	Slightly Dissatisfied	Neutral	Slightly satisfied	Satisfied	Very Satisfied

	20. Food and Beverage choices 21. Efficiency of Check-in staffs 22. Friendliness of staff 23. Grooming and Appearance of staff 24. Network of destinations / Connectivity 25. Innovative marketing schemes like Special pricing, Holiday packages, etc 26. Overall Satisfaction with this Airlines							
	27. Leg room available on seat 28. Quantity of Food given	Very Limited	Limited	Somewhat Limited	Neutral	Somewhat Plentiful	Plentiful	Very Plentiful
	29. Food Serving Experience	Very Poor	Poor	Below Average	Average	Above Average	Good	Excellent

	30. Special facilities of the Airlines for Children, the Elderly and the Handicapped							
	31. Seating Comfort	Very Uncomfortable	Uncomfortable	Somewhat Uncomfortable	Neutral	Somewhat Comfortable	Comfortable	Very Comfortable
	32. Smooth Take-off / Landing	Very Rough	Rough	Somewhat Rough	Neutral	Somewhat Smooth	Smooth	Very Smooth
	33. Staff Assistance at the time of Arrival	Very Unhelpful	Unhelpful	Somewhat Unhelpful	Neutral	Somewhat Helpful	Helpful	Very Helpful
	34. Staff Attitudes at the time of Check-in	Very Negative	Negative	Somewhat Negative	Neutral	Somewhat Positive	Positive	Very Positive

Facilities & Services during Flight	1. Cleanliness of Check-in Counter 2. Cleanliness of Lounge 3. Cleanliness of Washrooms 4. Conditions of Aircraft Carrier 5. Internet / WIFI / sky phone options during Flight 6. Lounge Comfort and Space 7. Luggage delivery times 8. Medical Facilities available during Flight 9. Newspaper and Magazines service during Flight 10. Cleanliness of Cabin	Very Poor	Poor	Below Average	Average	Above Average	Good	Excellent
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Communication & Information	1.Communication skills of Staff 2.Dissemination of Information in case of Delay of Flight 3.On-time Flights departure and arrivals 4.Personalized attention given to you 5.Post-service Relationship 6.Security of the Luggage	Very Poor	Poor	Below Average	Average	Above Average	Good	Excellent
	1. Would you like to travel again with this Airline? 2. Would you Recommend this Airline to others	Yes	No					

Open-ended questions: Your name, Age, Which Airline would you prefer and why? & Any Additional comments or suggestions